



Ti5 Metasystem Manual

Release 2.0.6

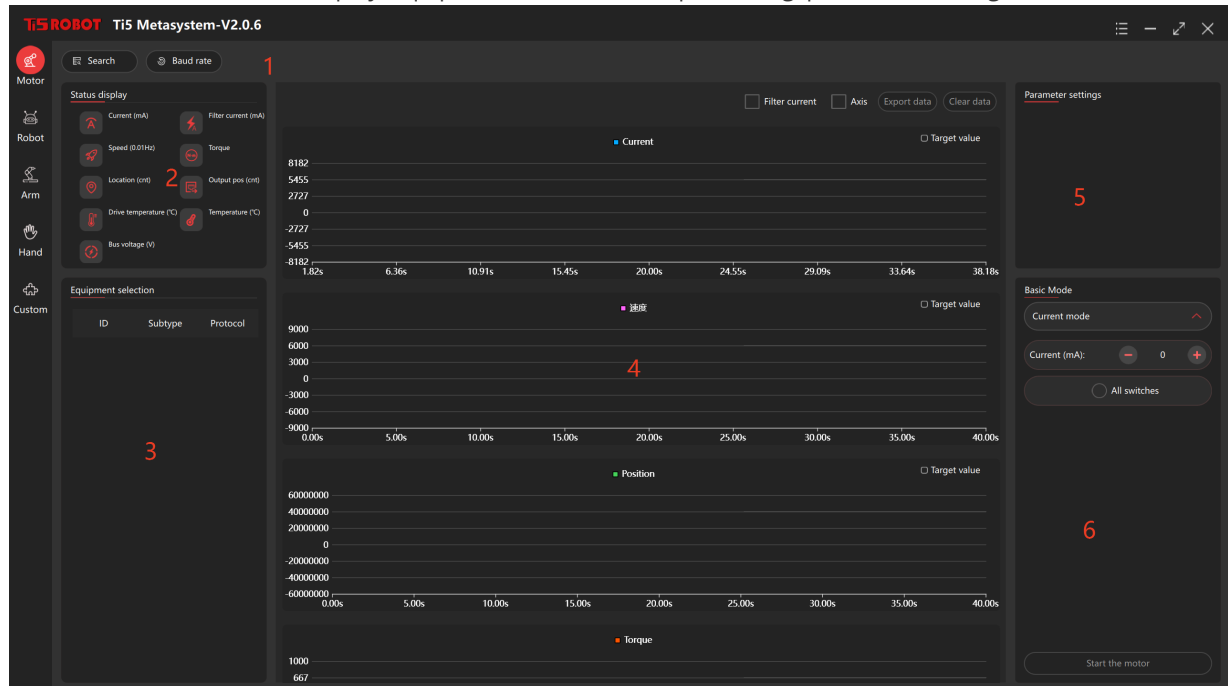
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1. Motor Control Section

Pre-Use Instructions

Before powering on, ensure the supply voltage does not exceed 48V. The motor control page consists of 6 modules: menu bar, status display, equipment selection, data processing, parameter setting, basic mode.



1.1 Menu Bar

The menu bar has device search and baud rate detection buttons.

1.1.1 Equipment Search

Click on "Device Search" and the Device Search pop-up window appears. After selecting the search parameters, click the "Search" button to start the search, and then the motor searched will be added to the equipment selection list. Search parameter explanation: Equipment number refers to the equipment number of Can analyzer, which defaults to 0 according to the insertion sequence. The equipment number +1 is inserted for each additional analyzer. After the analyzer with equipment number 0 is unplugged, the equipment number 1 before becomes 0. Communication mode refers to the communication protocol of motor, which includes CAN and CANopen protocols. Channel selection refers to the channel on Can analyzer. Baud rate refers to the baud rate of analyzer communication. Timeout time refers to the maximum waiting time for each communication, in milliseconds. When the communication is unstable, the timeout time can be increased to ensure the stability of communication. Search ID range, search motor according to start ID and end ID, reduce unnecessary search range and improve efficiency.

Search for motor

×

Device number

0

^

Protocol

Can-CXKJ

^

Channel

Channel 1

^

Baud rate

1000Kbps

^

Timeout period

−

100

+

Search ID Range

StartID

−

0

+

CloseID

−

10

+

Search

(Tips: Please use the ChuangXin Technology Supreme Analyzer.)

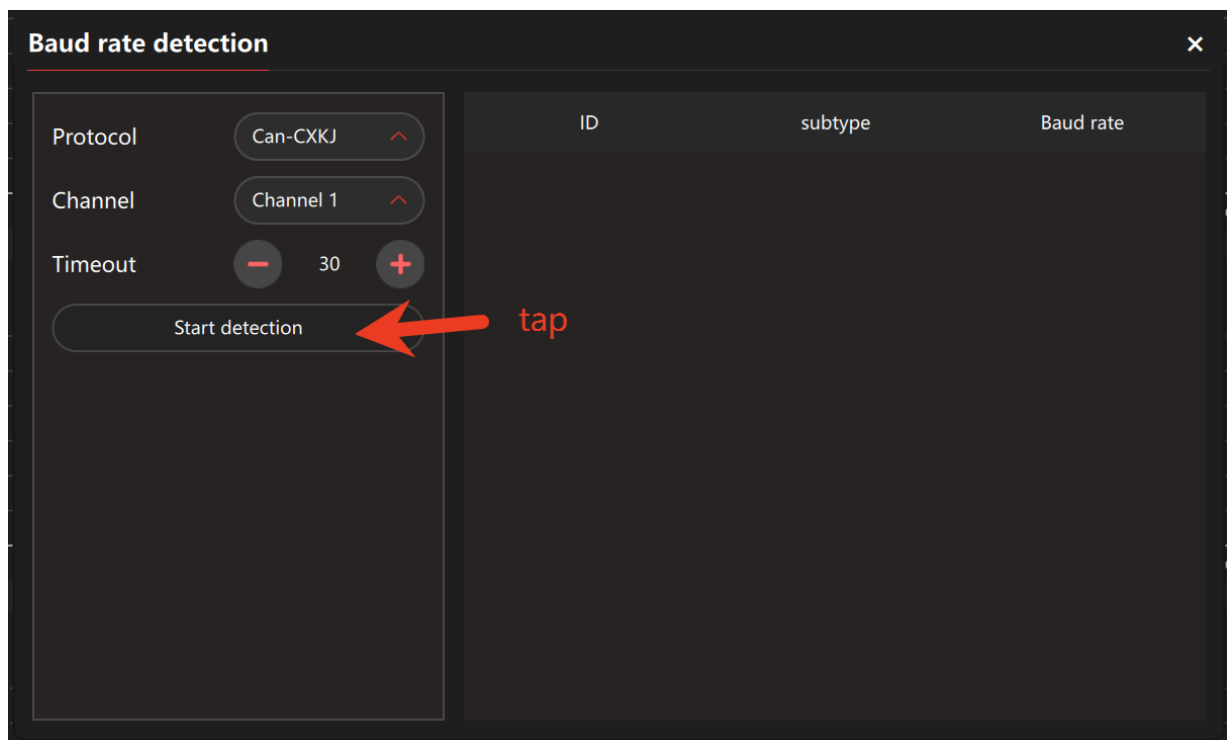
Confirm

tap

1.1.2 Baud Rate Detection

Click on "Baud rate detection" to display the Baud rate detection pop-up window. After selecting the detection parameters, click the "Start Detection" button to start detection, and then the detected motor will be added to the list on the right, click "End Detection" to stop this detection.

Explanation of detection parameters: communication mode refers to the communication protocol of the motor, there are two protocols of CAN and CANopen; channel selection refers to the channel on the Can analyzer; timeout time refers to the maximum waiting time of each communication, the unit is milliseconds. When the communication is unstable, the timeout time can be increased to ensure the stability of communication.



1.2 Status Display

After starting the motor, parameters such as current, speed, position, bus voltage, drive temperature, and motor temperature will be displayed with a refresh rate of 240 ms.

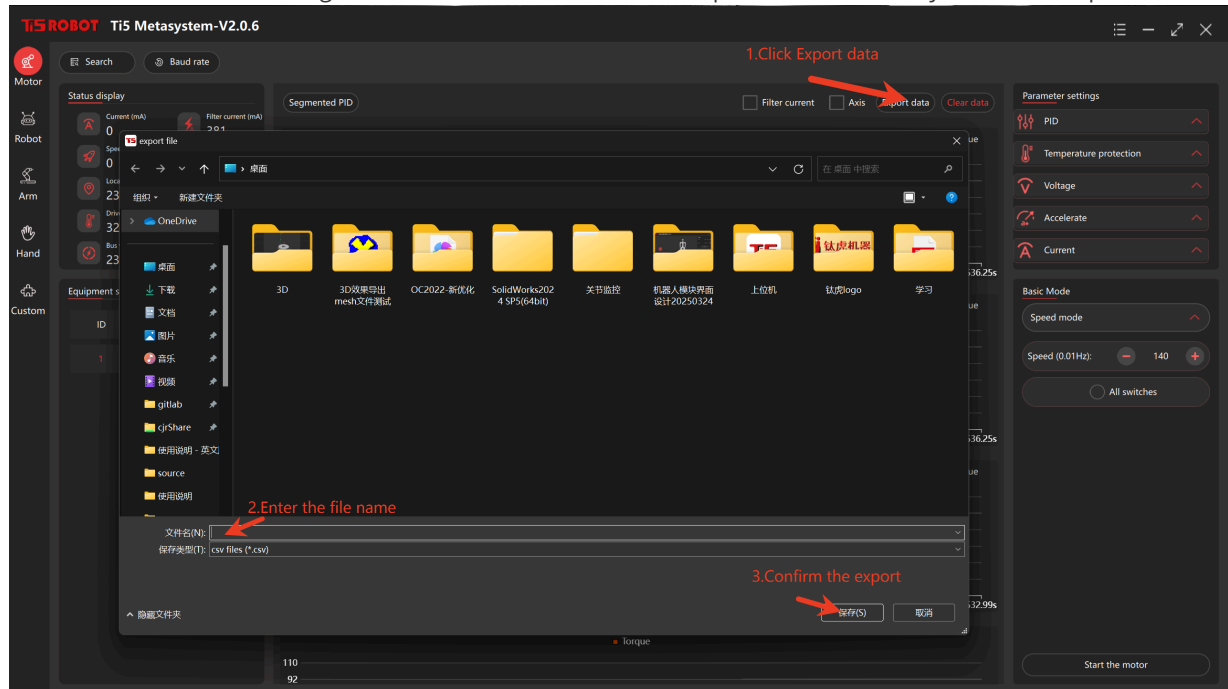
1.3 Equipment Selection

After the search is completed, the searched motor will be added to the lower list. Click the selected motor with the mouse to obtain the parameters of the motor and the subsequent starting motor for data acquisition and working mode control.



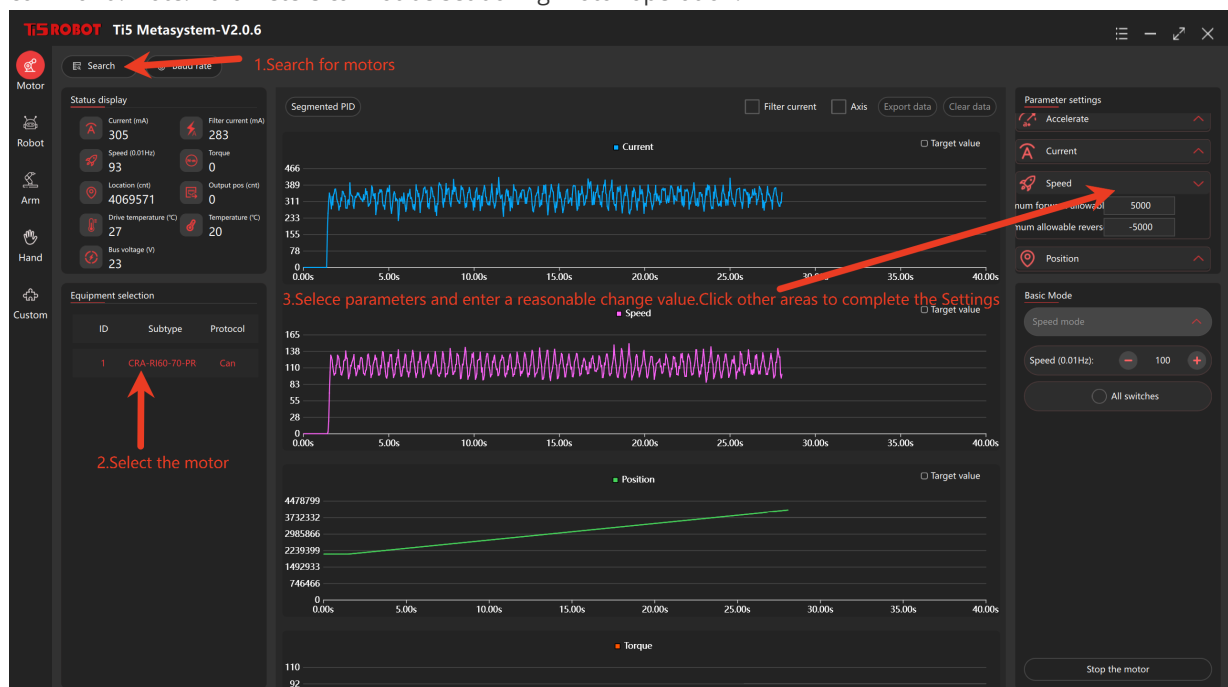
1.4 Data Processing

After starting the motor, the waveform diagram displays parameters such as current, speed, position and torque of the motor, updated every 40 ms. When the filtered current at the top is selected, the filtered current will be displayed in the current waveform diagram at the bottom. When Axis is selected, the line on the horizontal axis in the waveform diagram will be displayed. After closing the motor, export data and clear data buttons are enabled. Click "Export Data" to pop up the file pop-up window. Select the folder and enter the saved name. After confirmation, the motor data collected in this operation can be exported. Click "Clear Data" to clear the data in the waveform diagram and the data collected in this operation recorded by the host computer.



1.5 Parameter Setting

When the motor is selected, the parameters of the motor will be displayed in this column. After entering reasonable parameters in the parameter input box, click outside the input box or enter to execute the setting command. Note: Parameters cannot be set during motor operation.



1.6 Basic Mode

Control the collection and working mode of motor parameters. Select the working mode from the drop-down box, click "Start Motor", the mode cannot be changed, start collecting data, and display it in the waveform diagram and status display column; input the working mode parameters, enter reasonable parameters in the input box, click Enter, and then send them to the motor, and the motor starts to run; if you want to control all motors synchronously at the same time, select "All Switches" before starting the motor, and you can control synchronously. If the motor reports an error after startup, a pop-up window will display the error type and code.. It is necessary to clean up the error before controlling the motor.



2. Robot Control Section

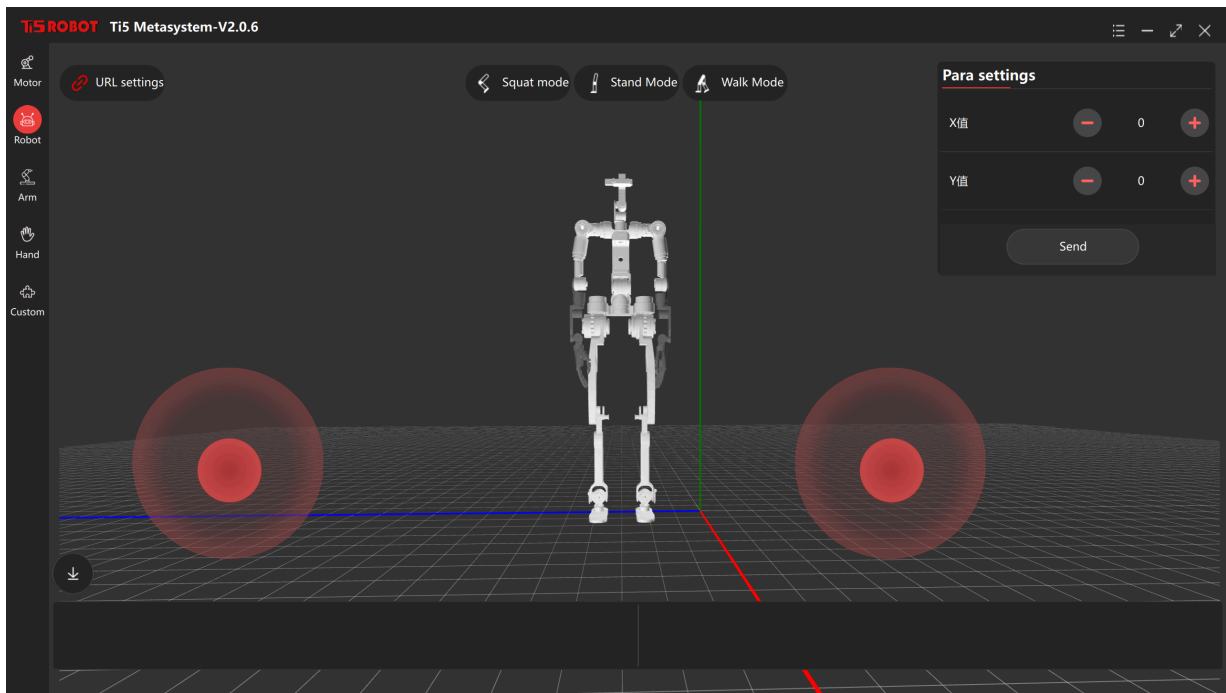
Pre-Use Instructions

Before starting, make sure that the following have been completed:

- **Robot Hardware:** Make sure the robot is properly installed and connected to the control board.
- **Network connectivity:** Ensure that the computer and robot are on the same network and that network communication is normal.
- **Server applications:** You need to make sure that the robot's server applications are properly installed and running.

It is mainly divided into the following areas:

- **Connection area:** Used to configure and connect robotic devices.
- **Motion control area:** used to control the robot's movement, grasping and placement operations.
- **Status display area:** displays the robot's current status, sensor data, and motion feedback.
- **Log and prompt area:** used to display operation log and system prompt.



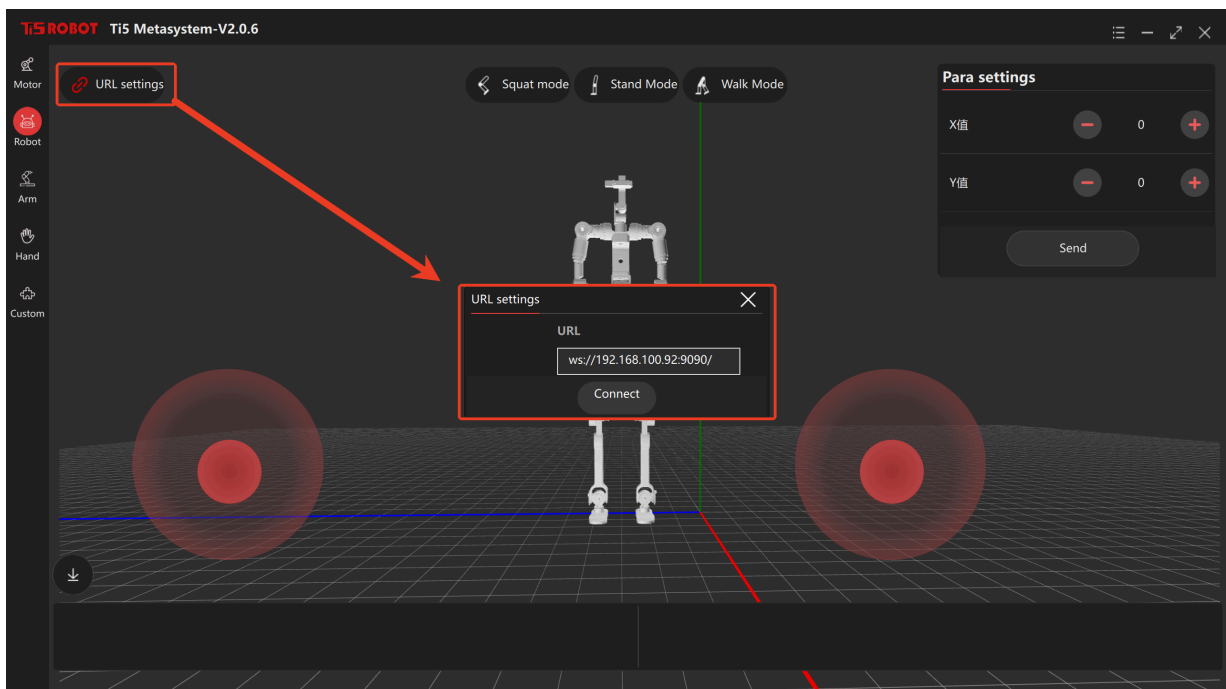
2.1 Network Connections

2.1.1 Start The Server Program

The user needs to start the server on the control board, start listening on the specified port, and provide the local IP address.

2.1.2 Set Network And Port

Click the URL Settings button, fill in the IP address and port number of the robot in the connection settings area of the pop-up window, and ensure that the input address is consistent with the address bound to the server program. Click the "Connect" button. If the connection is successful, the icon will turn green; if the connection fails, there will usually be an error prompt on the interface. The user needs to check the corresponding server status.



2.2 Motion Control

2.2.1 Knee Bending Mode

Click the "knee bending mode" button, and the robot moves or stays stable in a knee bending posture.

2.2.2 Straight-Leg Mode

Click the "Straight Leg Mode" button and the robot stays standing with straight legs.

2.2.3 Travel Mode

Click the "Walk Mode" button and the robot enters the walk mode.

2.2.4 Control Walking Mode

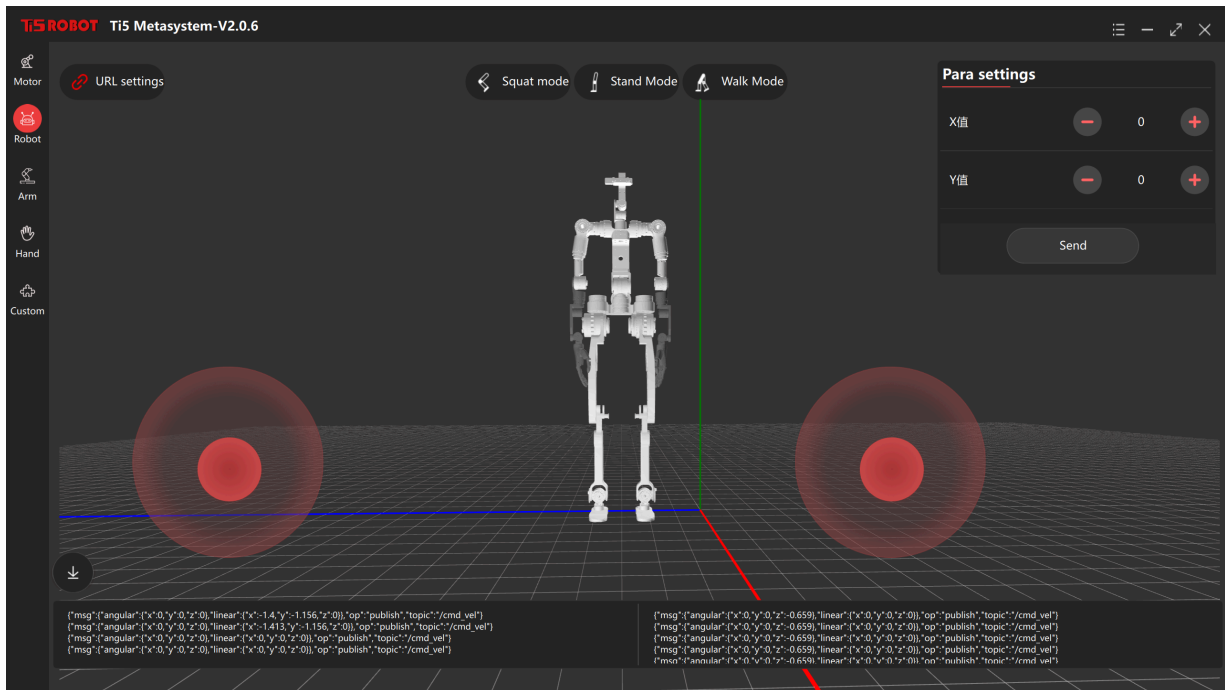
After switching to walking mode, the user can control the robot's motion and steering simultaneously through the virtual joystick.

	rocker area	moving direction	Movement amplitude/control speed	implication
motion control	Left circle	up/down	Distance from the center of the rocker to the current drag position	Forward/backward translational motion of robot
		left/right	Distance from the center of the rocker to the current drag position	Robot left translation/right translation
steering control	Circle on the right	left/right	Distance from the center of the rocker to the current drag position	Robot turns left/right

2.2.5 Custom Motion Direction Mode

X, Y value range (-1.5,1.5), click the "Send" button after confirming.

2.3 log Viewing



View Log button in lower left corner to view messages sent. Click the button,"arrow down" to close the log box,"arrow up" to display the log box.

3. Robot Arm Control Module

Environmental Preparation

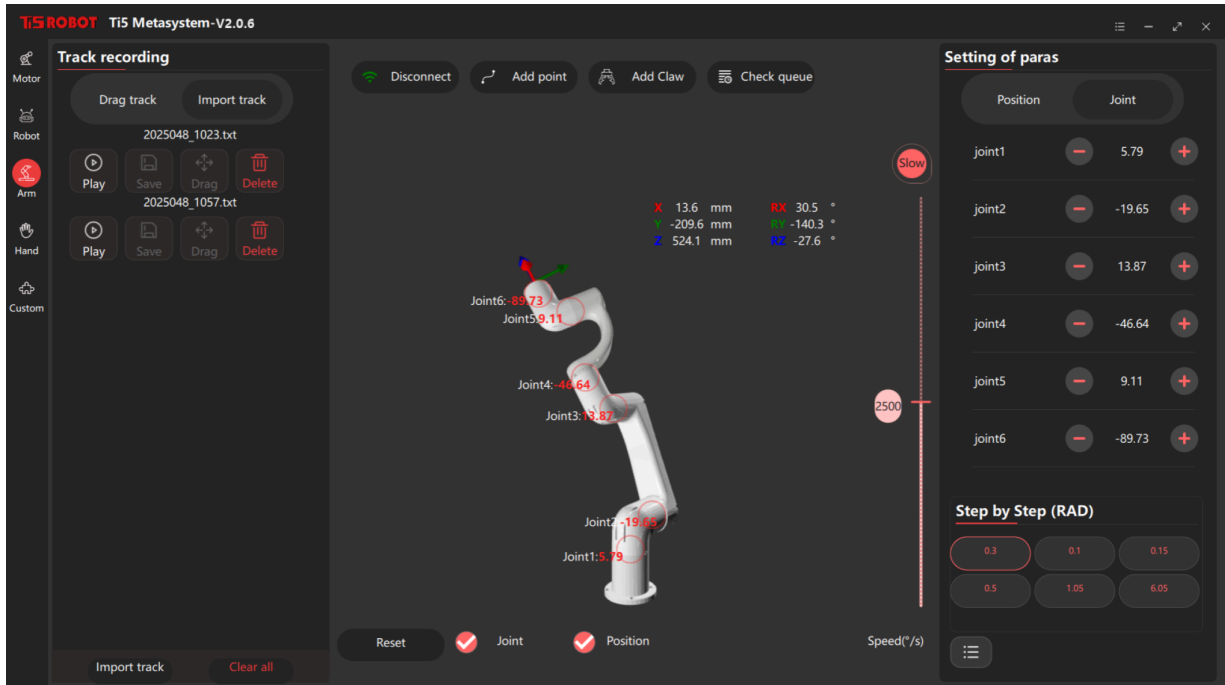
1. Ensure that the power supply is 24V and the current is 13A.
2. Connect the robot arm and power cable, resistor, CAN analyzer (LINUX version). Connect IPC and USB cable.
3. The tablet and industrial computer are connected to the same network terminal.
4. Do not plug and unplug wires at will during equipment operation.

3.1 Device Attachment

Click the "Connect" button, fill in the IP address of the corresponding connection, and select the weight of the robot arm. Click the "Confirm" button, the WIFI icon turns green, indicating that the network connection is successful, and the CAN analyzer is connected successfully. The icon turns red. Please check whether the power supply is powered on, whether the CAN analyzer is connected successfully, and whether the robot arm and the industrial computer are connected to the same network terminal.

3.2 Joint Space Motion

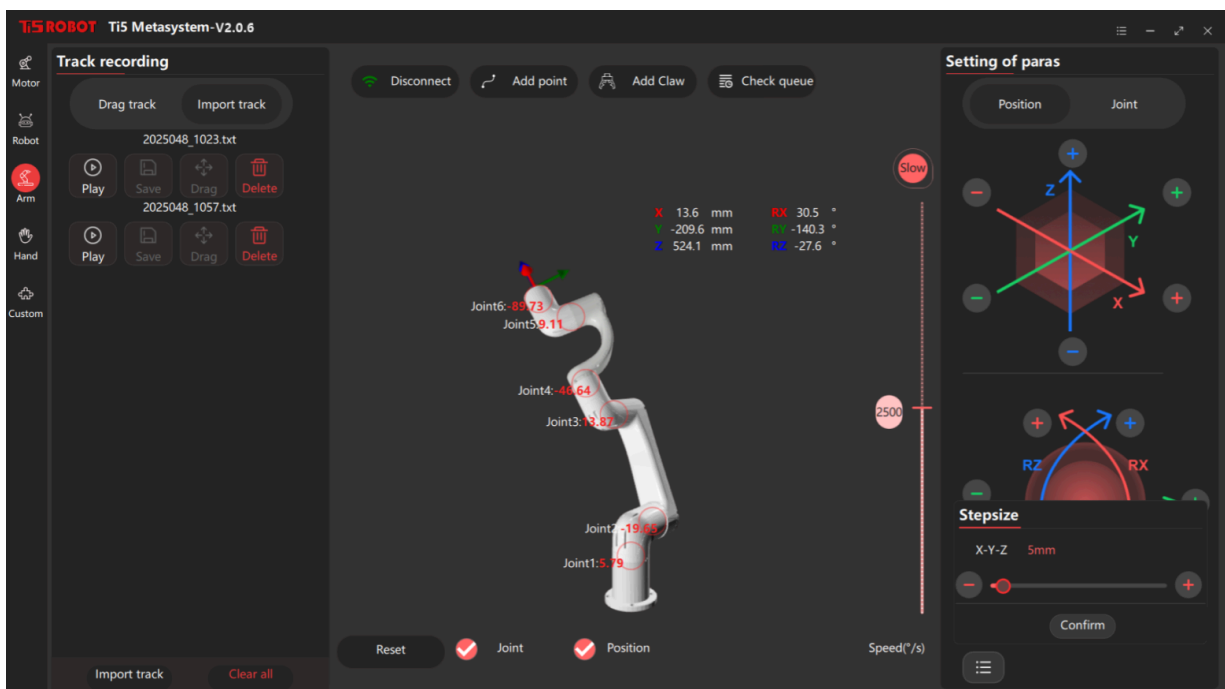
This mode allows you to control the rotation of a single joint of the robot arm.



1. Click the button below, select the rotation step value of each joint of the robot arm (RAD radian system), and adjust the step size of the robot arm movement.
2. Select the modified manipulator joints 1 - 6, click "+" to rotate outward, click "-" to rotate inward. After releasing, the interface displays the corresponding coordinates of the robot arm.
3. Move the speed slider bar to modify the speed of the robot arm.

3.3 Cartesian (Position) Spatial Motion

When in this mode, controls the end of the robot arm to move along and rotate about XYZ axes in the tool coordinate system.



1. Click the button below to select the robot end motion to move the axis step size in XYZ direction (mm mm).

2. Refer to the arrow direction of the interface and click the "+" button or "-" button to select the direction of modification. After releasing, the interface displays the corresponding coordinates of the robot arm.
3. Move the speed slider bar to modify the speed of the robot arm.

3.4 Drag Teaching

This mode allows you to drag the robot arm and record the trajectory of the movement.



1. Drag Function

Each time you click the Drag button, when the button changes to End, start dragging freely. When you finish dragging, click End.

2. Play Function

Click Play to automatically play the track you just dragged.

3. Preservation Function

Click "Save" to name the dragged track record with the current time and store it for later viewing.

4. Delete Function

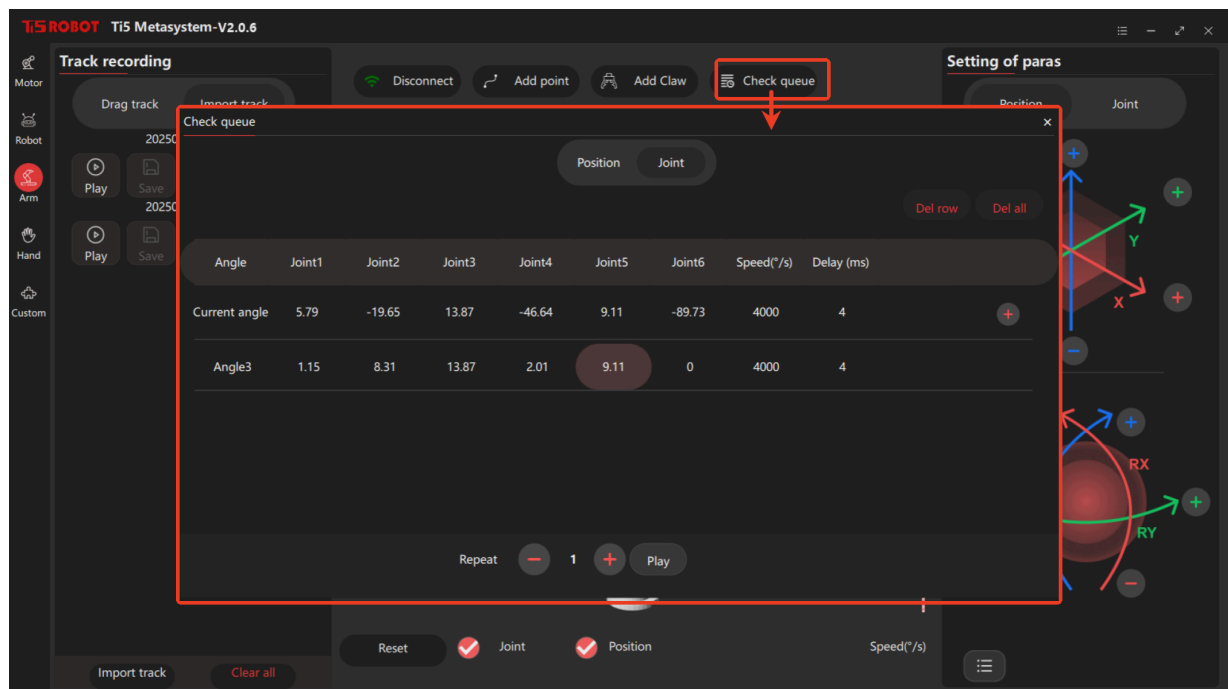
Click "Delete" to delete the corresponding track.

5. Import Function

Switch to the Import Tracks bar, and in case of a successful connection, all previously saved files will be imported. After importing tracks, there are also "Play" tracks and "Delete" functions.

3.5 Custom Motion

This mode is upgraded during the movement of the above three modes. By adding the points you want to reach, you can independently plan the movement route of the robotic arm.



1. Add Points

Click the Add Point button to add the current point. Or after modifying the information in the first line of the list, click the "+" button to add the current point.

2. View List

Click the View List button and select Joint mode to see the current 6 joint angle values, as well as the speed of the motion and the delay milliseconds for each action.

3. Modify The List

Double-click the cell, select the data you want to modify, modify and press Enter.

4. Delete Corresponding Points

Click any cell of the corresponding row, select Delete Row, and delete the corresponding row. Or the "Delete All" button to delete all points.

3.6 Observation Robot

This mode provides a robot model with coordinates, zoom in and out by mouse scrolling, left button following rotation, right button moving position, convenient for operator observation and understanding.

1. Display Joints

Click "Display joint values" to display the 6 corresponding joint values (degrees) of the robot arm, which cooperate with the joint space movement and follow the movement.

2. Display Position

Click "Display Position Value" to display the coordinates (mm, degrees) of the end of the robot arm, with Cartesian (position) space movement, follow the movement.

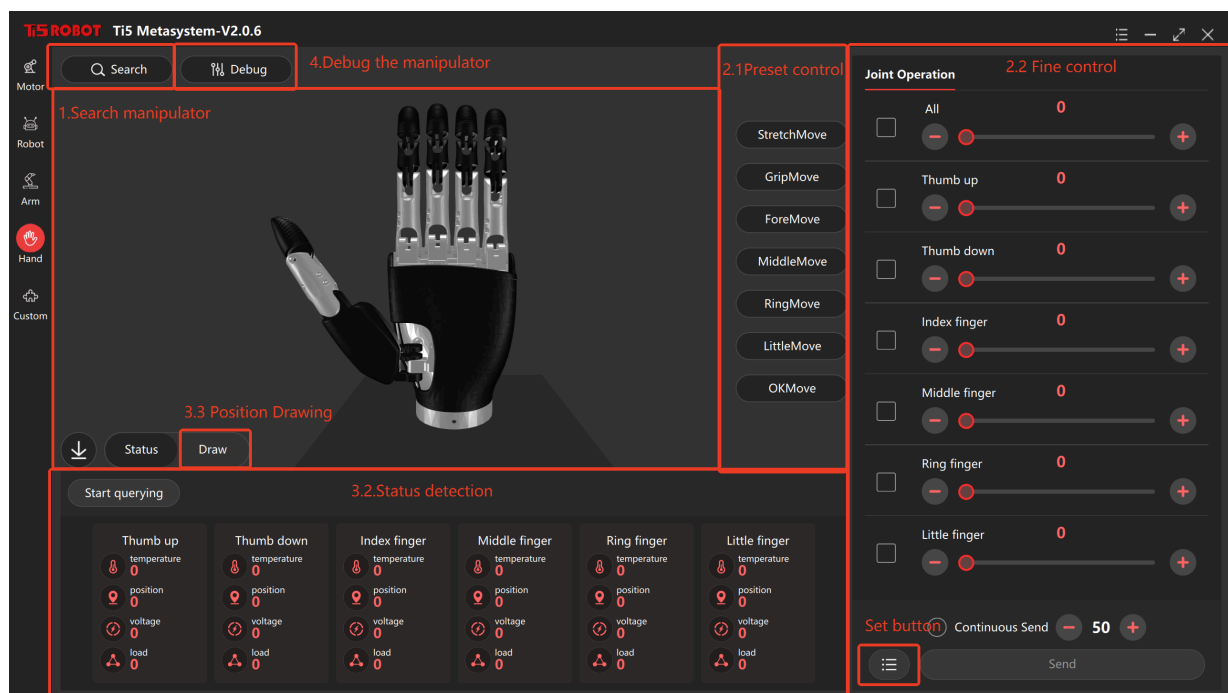
3. Return To The Original Point

Click "Back to Origin" to return the robotic arm to the initial state set by the developer.

4. Manipulator Control Section

Quick Start Guide

Before power-on, make sure that the supply voltage is set to 7V and the current is set to 3A.



4.1 Search Robot

Turn on the PC, and the first step is to find the search button and click it. The robot supports RS485 and CAN protocol communication. After selecting the required communication mode by switching the switch, click Search to get the corresponding "serial port"(get "id" when CAN communication). Click "Open" to connect the robot.

4.2 Control The Manipulator

4.2.1 Preset Control

There are seven preset actions on the right side of the model display area, such as "stretching", "clenching" and "extending index finger". After successful connection, the corresponding control can be realized.

4.2.2 Fine Control

There is a "Joint Operation" box on the right side of the preset action. You can adjust the angle value of the corresponding joint through the slider bar. Click "√" for the joint to be adjusted, and you can send it through the "Send" button at the bottom. When sending, if the option of "Continuous sending" is checked, the system will automatically send the joint information marked with "√" every once in a while (default 50ms).

There is a setting button on the left side of the send button. Open it to adjust the finger control mode. The functions of each mode are as follows.

Finger Position Mode	Pattern specification
Finger mode (default)	One command only operates one joint, including the position, acceleration, speed, torque and other information of the joint.
Position mode	One command operates all joints, including only the position information of each joint
Acceleration pattern	One command operates all joints, including only acceleration information of each joint
Speed mode	One command operates all joints, including only the speed information of each joint
Torque mode	One command operates all joints, including only torque information for each joint

4.3 Observation Manipulator

4.3.1 Reduction Of Models

The model display area has a real restoration model of the manipulator, which can be used to observe the posture of the manipulator. The model continues to synchronize the robot state after the robot is connected.

4.3.2 Status Detection (RS485 only)

Physical parameters such as "temperature", "position", "voltage" and "load" of the motor are displayed below the display area. After clicking Start Query, the parameters will be updated.

4.3.3 Location Mapping

The toggle button at the bottom of the display area switches to "Draw" to see the position drawing chart, which is used to observe the change of each joint position over time. Click Start Drawing, and the system will query and display the shutdown position information. Check the grid button on the right to display vertical axes.

4.4 Debugging Manipulator

This function is only available when "RS485" is connected. Click the Debug button at the top of the display area. Advanced parameters of manipulator can be modified on this page. Switch switch button to "ON" before modification. Click the "Power Up and Down" button on the right to power on and power down. The green button indicates that the state is powered on at this time, and the red button indicates that the state is powered down.

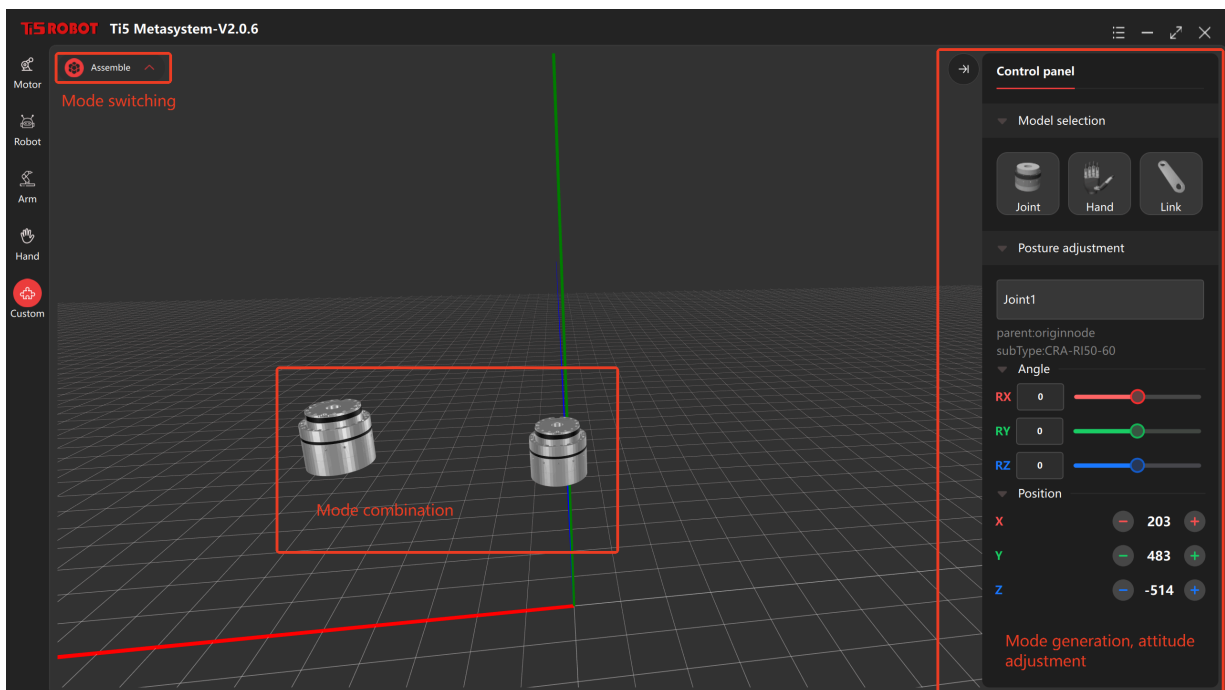
5. Custom Section

Pre-Use Instructions

There are two ways to use custom models, one is to build models in build mode and then switch to control mode, and the other is to load models directly through the right menu bar "Load" option.

In the main viewing area, you can hold down the left mouse button and drag to change the camera angle of view. Enter "W, A, S, D, R, F" on the keyboard to change the camera position in space.

5.1 Model Construction



5.1.1 Entering Build Mode

Switch the custom module operation mode to "Build Mode" via the Mode Switch button. The basic material library provided in the construction mode is divided into three categories: "motor", "robot" and "connecting rod".

5.1.2 Model Addition

From the right gallery, press and drag a model to the main viewport to complete the creation. Right click the mouse above the model to call out the menu bar for "Copy","Replace","Delete" and other operations.

Copy operation, you can right-click in the blank out of the menu bar to select paste.

After replacement operation, you can select a model in the pop-up model selection interface. Currently, 7 motor models, 2 manipulator models and 5 linkage models are supported.

Delete operation has the option of "Include whole chain", that is, when there is a model connected to this model as a child model, delete with whole chain will be deleted together.

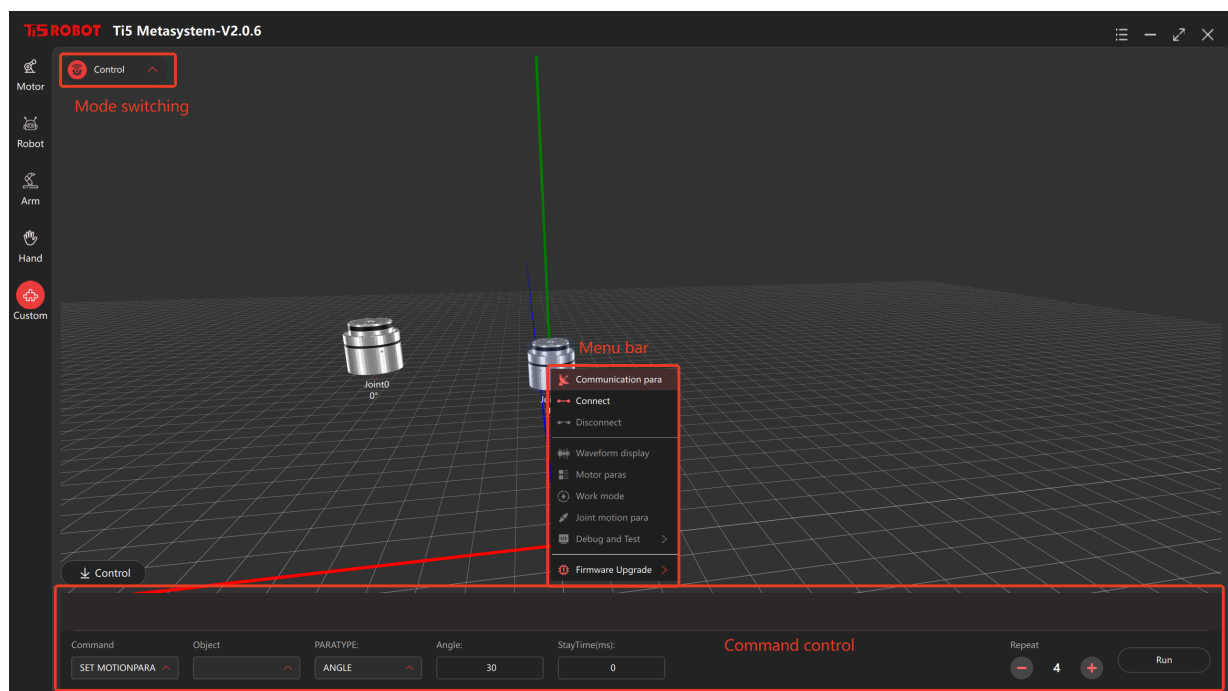
5.1.3 Model Combination

When more than one model has been generated, you can hold down one model and drag the mouse over another to locate the target model. The selected model is identified in blue and the target model in green.

The models are connected in the default interface mode, and the rotation angle and offset distance can be adjusted through the attitude adjustment bar. You can also adjust the name of the model in this column.

Note: The drive link can only be used as a child model of the motor model, it does not rotate with the motor itself, and transmits the motor rotation of the parent model to the turntable of the model.

5.2 Model Control



5.2.1 Entering Control Mode

Switch the custom module operation mode to "Control Mode" through the mode switch button.

5.2.2 Right-Click Menu Bar

Right click on the motor model menu bar as follows

Menu Bar Option Name	Option function
Communication para	Can set motor communication parameters, mainly ID to fill in correctly

Menu Bar Option Name	Option function
Connect	Connect the motor. When the connection is successful, the motor information is displayed in green.
Disconnect	Disconnected
Joint motion para	Angle, forward and reverse angle limit, forward and reverse speed limit parameters can be set
Set zero angle	Set current position to zero
Waveform display	Open waveform display window to display current, speed, position and torque parameters
Motor paras	View and edit PID, acceleration, current, velocity, position and basic information
Piecewise PID	Set PID related parameters
Work mode	Motor can be controlled in different modes
Response time test	Response time testing for all motor modes

When right-clicking the robot model, the menu bar is as follows:

Menu Bar Option Name	Option function
Communication para	Search Robot
Connect	Link Manipulator
Disconnect	Disconnected

Right click on the blank menu bar as follows,

Menu Bar Option Name	Option function
Search all	Search for all connectable motors
Connect all	Connect all devices that exist in the space
Disconnect all	Disconnect all connected devices

Menu Bar Option Name	Option function
Clear alarm	Clear alarm
Set zero angle	Set all motors to zero
Save as	Save the model and its communication parameters
Load	Load an existing model

5.2.2.1.1 Setting Communication Parameters

Before connecting the motor, configure the parameters such as equipment number, channel number and baud rate. When setting communication parameters, if "Auto Match CAN ID" is enabled, it can be automatically matched when running "Connect All" regardless of device number and channel.

Set communication parameters Joint8 [X]

Equipment number: 0

Protocol: Can-CXKJ

Channel: Channel 1

Baud rate: 1000Kbps

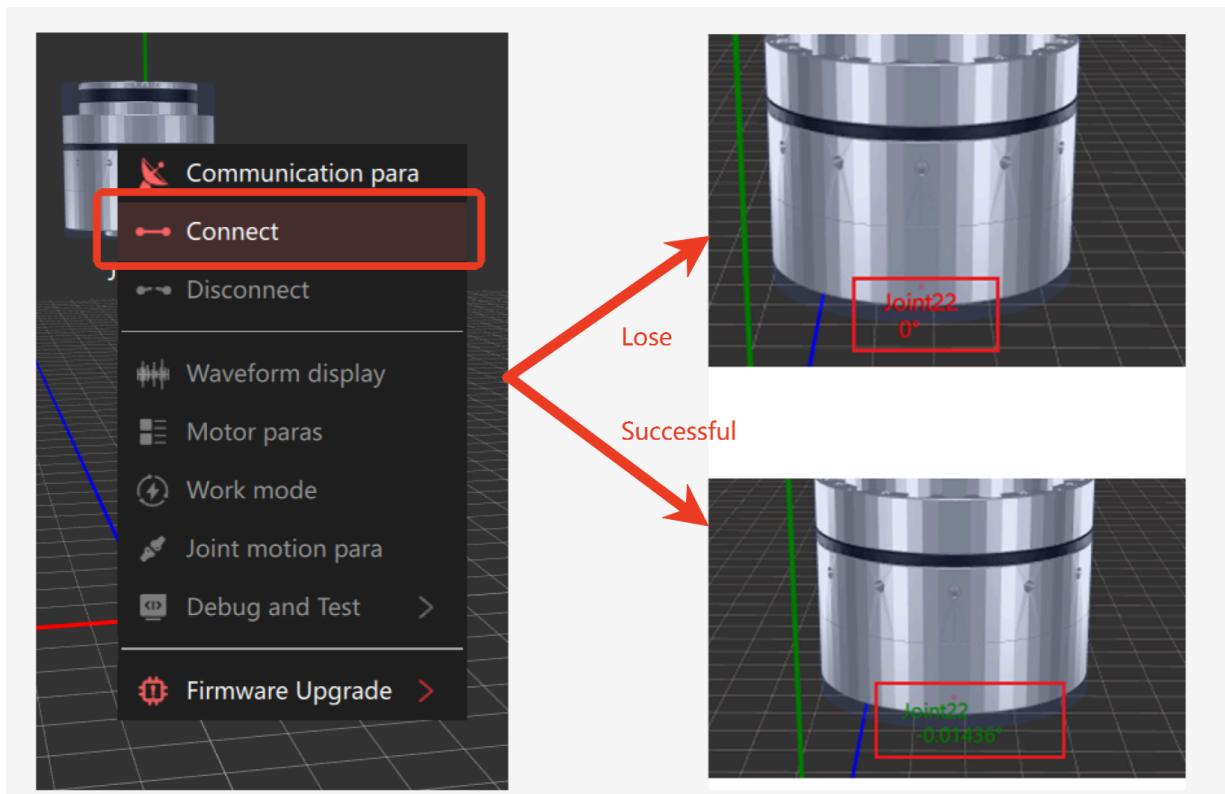
CAN ID: 0

Auto match CAN ID: On

Setting

5.2.2.1.2 Connecting The Motor

After configuring the communication parameters, click Connect to connect the motor. If there is a connection failure, check whether the communication parameter configuration is correct.

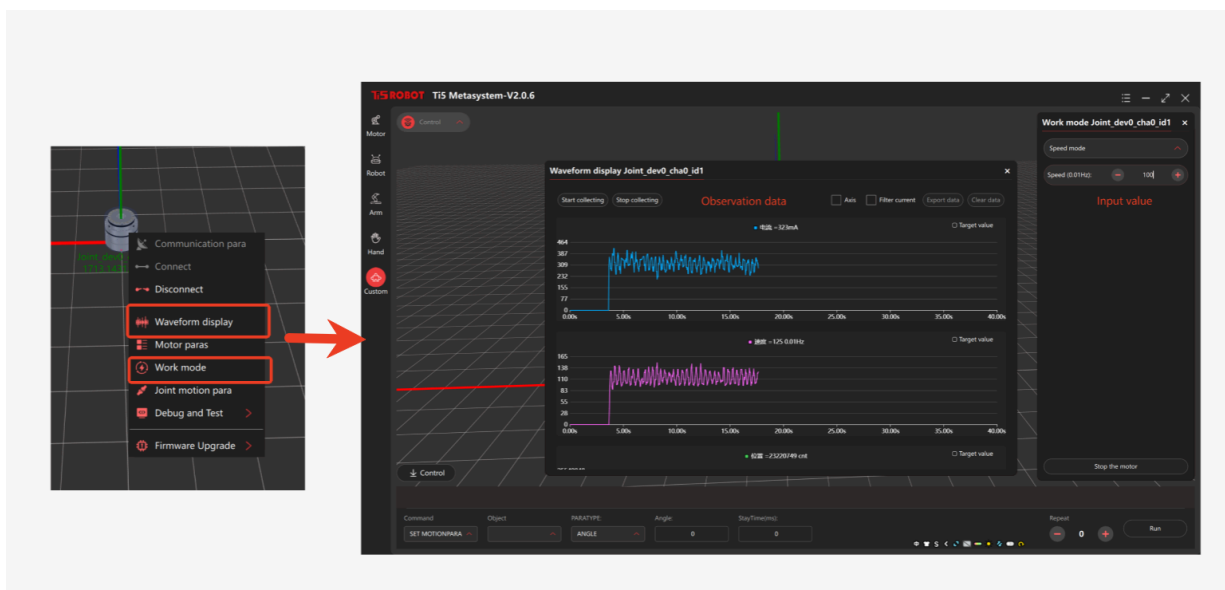


5.2.2.1.3 Mode Of Operation

The working mode includes current mode, speed mode, position mode, output position mode and torque mode. It can be operated by directly inputting the numerical value.

5.2.2.1.4 Waveform Display

Using "waveform display" needs to be used in conjunction with "working mode", we can observe the current, speed, position and torque of the motor in the waveform diagram.

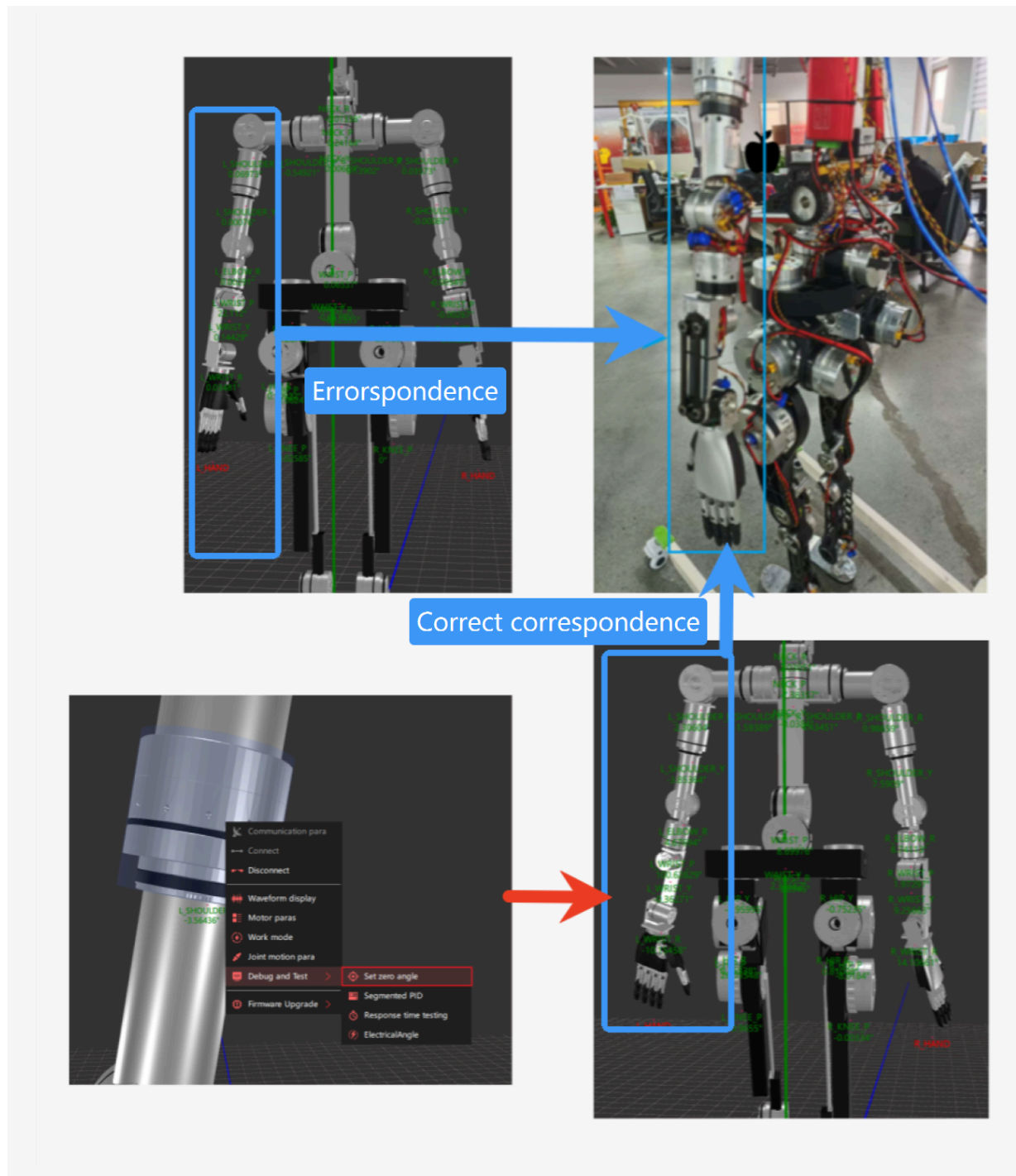


5.2.2.1.5 Setting Zero

Zero logic means that when every motor of the physical robot enters a natural posture, the robot represented by the model does not correspond, because the angle value fed back is not zero. In the face of this kind of time, we should manually break or control the motor to enter the zero position state that we think, and then set the zero position for the corresponding motor to achieve the effect of the model and the object.

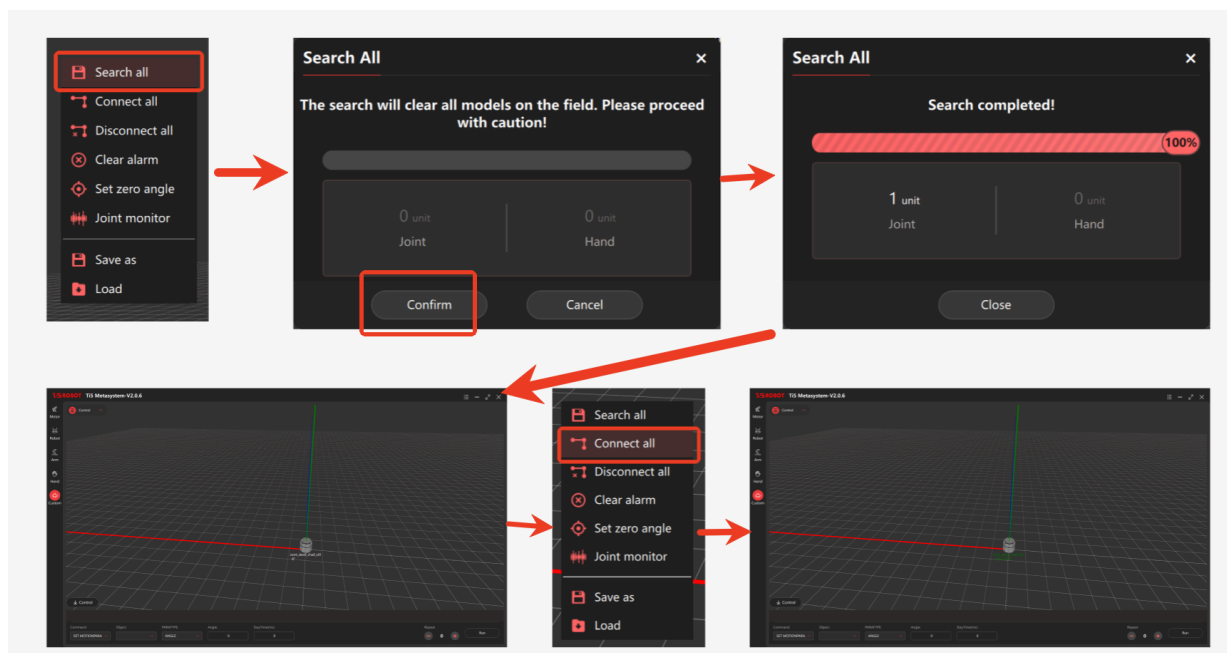
Knowing the logic of zeroing, you can use the "Set Zero" button in the right-click menu in the blank. The zero setting of this entry will set all connected models, make sure that each motor is in the correct position before use.

As shown in the figure below, the robot arm poses correctly, but the feedback data in the model shows abnormality (eversion). We know through logical analysis that the motor "L_SHOULDER_Y" did not find the correct zero. At this point we find the problem motor right click and select "set zero", you can solve the problem.



5.2.2.2.1 One-Click Search

"One-click search" function can automatically help you search the motor and configure the communication parameters when you do not know the motor communication parameters. As shown in the figure below, we need to connect 31 motors and 2 manipulators at the same time. If we create models one by one and configure communication parameters one by one, it will be a bit cumbersome. Use the one-button connection function to automatically complete the steps of creating models and configuring communication parameters.



5.2.2.2 Connect All

When using Connect All, if a motor selects "Automatic Matching" on the communication configuration interface, the motor will use the automatic matching algorithm to find the CAN index and channel corresponding to its id.

5.2.3 MML Command Control

The area below the main view area is the command input area and the command visual input area. By selecting MML (Man Machine Language) command + operation object + parameter, the control of joint/manipulator can be realized.

5.2.3.1

MML commands supported by joint motors include the following:

order	function
SET RUNMODE	Set the operating mode
SET DEVICEPARA	Set device parameters
SET CURRENTPARA	Set current parameters
SET SPEEDPARA	Set speed parameters
SET POSITIONPARA	Set Position Parameters
SET SYSCMD	Set system command parameters
SET COMMUPARA	set communication parameters

order	function
SET MOTIONPARA	Set motion control parameters

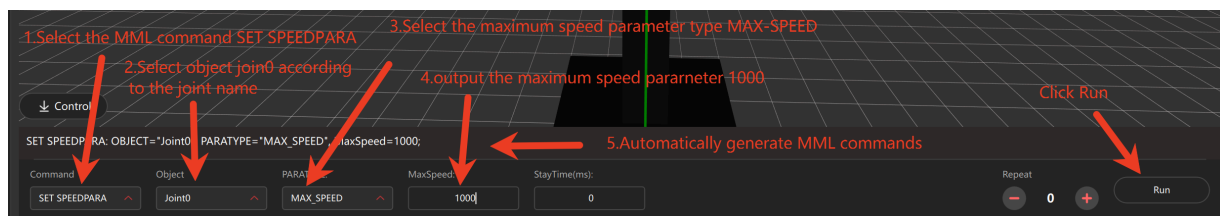
5.2.3.1.1 Preparation Before Operation

To control using MML commands, ensure the joint motors/manipulator are in a connected state. Please refer to the relevant chapter for operations.

5.2.3.1.2 Operation Steps

The following demonstrates the operational steps for configuring motor speed parameters using MML commands:

- step 1. COMMAND Select the MML command SET SPEEDPARA.
- step 2. OBJECT Specify the target joint (e.g., Joint0) as the operation object.
- step 3. PRARTYPE Choose the parameter type (MAX_SPEED for maximum speed).
- step 4. MaxSpeed Enter the desired speed value (e.g., 1000).
- step 5. The MML command script is automatically generated in the text box.
- step 6. Click Run to apply the maximum speed setting.



5.2.3.1.3 Set Runmode

Motion control of joint motors can be achieved using the Run Mode command. The following is a list of supported run mode commands:

RUNMODE	function	name of parameter	parameter meaning
SPEED	Set joint motor operating mode to speed mode	SPEED	Speed value of speed mode, converted to degrees per second Formula: (speed value/100/deceleration ratio)*360
CURRENT	Set joint motor operating mode to current mode	CURRENT	Current mode current value in mA
POSITION	Set joint motor operating mode to position mode	POSITION	The cnt value of the target position, converted to the reducer angle formula: (position value (cnt)/65536/reduction ratio)*360
STOP	Set joint motor operation mode to stop mode	NA	

5.2.3.1.4 Set Equipmnt Parameters Set Devicepara

Use the Equipment Parameters command to set the equipment parameters of the articulated motor. The following is a list of supported device parameter commands:

PARATYPE	function	name of parameter	parameter meaning
MAX_MOTORTEMP	Set joint motor temperature protection value	MaxMotorTemp	Motor temperature protection value in °C
MAX_PCBTEMP	Set joint motor motherboard temperature protection value	MaxPCBTemp	Motor motherboard temperature protection value, unit: °C
MAX_VOL	Set maximum voltage	MaxVoltage	Maximum voltage in V
MIN_VOL	Set minimum voltage	MinVoltage	Minimum voltage in volts

5.2.3.1.5 Set Current Para

The current parameter command allows configuration of joint motor current settings. Below is a list of supported current parameter commands

PARATYPE	function	name of parameter	parameter meaning
MAX_CURRENT	Set maximum positive current	MaxCurrent	Maximum positive current in mA
MIN_CURRENT	Set minimum negative current	MinCurrent	Minimum negative current in mA
PROPORTIONAL	Set current loop ratio	Proportional	current loop ratio
INTEGRAL	Set Current Loop Integral	Integral	current loop integral

5.2.3.1.6 Set Speed Parameter Set Speedpara

The speed parameter command enables configuration of joint motor speed settings. The following speed parameter commands are supported:

PARATYPE	function	name of parameter	parameter meaning
MAX_ACC	Set maximum positive acceleration	MaxAcceleration	Maximum positive acceleration, which is the allowable speed increase per millisecond

PARATYPE	function	name of parameter	parameter meaning
MIN_ACC	Set minimum negative acceleration	MinAcceleration	Maximum negative acceleration, which is the allowable speed increase per millisecond
MAX_SPEED	Set maximum allowable forward speed	MaxSpeed	Maximum forward speed, converted to degrees per second Formula: (speed value/100/deceleration ratio)*360
MIN_SPEED	Set minimum allowable negative speed	MinSpeed	Minimum negative speed, converted to degrees per second Formula: (speed value/100/deceleration ratio)*360
PROPORTIONAL	Set Speed Ring Scale	Proportional	P Parameter of PID Control of Velocity Loop
INTEGRAL	Set speed loop integral	Integral	I Parameter of PID Control of Speed Loop

5.2.3.1.7 Set Positionpara

The position parameter command enables configuration of joint motor positioning parameters. The following position parameter commands are supported:

PARATYPE	function	name of parameter	parameter meaning
MAX_POS	Set maximum forward position	MaxPosition	Maximum forward position cnt value, position value is (reducer target angle/360)reduction ratio 65536
MIN_POS	Set minimum negative position	MinPosition	Minimum negative position cnt value, position value is (reducer target angle/360)reduction ratio 65536
PROPORTIONAL	Set Position Ring Scale	Proportional	Position ring is commonly used PD control
DIFFERENTIAL	Set Position Ring Differential	DIFFERENTIAL	Position ring is commonly used PD control
Offset	Set Position Offset	Offset	Set offset value and target position (target position = encoder position-offset value)

5.2.3.1.8 Setting System Command Parameters Set Syscmd

System command parameters enable configuration of joint motor system settings. The following system commands are supported

PARATYPE	function
RESET_FAULT	clear alarm
STORE_FLASH	Save parameters to Flash

5.2.3.1.9 Setting communication parameters Set Commupara

He communication parameter command enables configuration of CAN communication parameters for joint motors. The following communication parameter commands are supported:

PARATYPE	function	name of parameter	parameter meaning
CAN_ID	Set CAN ID	CanId	CAN ID value for joint motors for CAN communication, range: 1 to 127
BAUD_RATE	Set CAN communication baud rate	BaudRate	CAN communication baud rate, valid values: 1000, 500, 250, 125, 100, 50, unit Kbps

5.2.3.1.10 Set Motionpara

Motion control parameters enable configuration of joint motor movement. The following motion control parameter commands are supported:

PARATYPE	function	name of parameter	parameter meaning
ANGLE	Set joint angle	Angle	Control the motor using position mode, where position value =(angle parameter set/360)reduction ratio 65536
REDUCE_RATIO	Set joint motor reduction ratio	ReduceRatio	Joint motor reduction ratio

5.2.3.2 MML Command Control Robot

The MML commands supported by the manipulator include the following, and the operation steps are similar to the joint motor control. Ensure that the manipulator is connected before operation. The following lists the supported MML commands:

order	function
SET HANDACTION	Set gestures

order	function
SET HANDPOS	Set finger pose
SET FINGERPOS	Set one-handed finger pose

5.2.3.2.1 Set Handaction

Gesture motion commands enable control of the manipulator to perform predefined gesture actions. The following gesture parameters are supported:

ACTION	function
OPEN	Action: Stretch gesture
CLOSE	Action: fist gesture
MOVE_INDEX	Action: extend index finger
MOVE_MID	Action: Extend middle finger
MOVE_RING	Action: Extend ring finger
MOVE_LITTLE	Action: Extend your little finger
OK_GESTURE	Action: OK posture

5.2.3.2.2 Setting Finger Pose Set Handpos

Finger pose commands enable individual manipulator fingers to achieve specified angular configurations. The following pose parameters are supported:

TYPE	function	Parameter 1 Name	Meaning of Parameter 1	Parameter 2 Name	Meaning of parameter 2	Parameter 3 Name	Meaning of parameter 3
ALLPOS	Set finger pose	LittleFinger	Pinkie pose angle	RingFinger	Angle of ring finger pose	MiddleFinger	middle finger pose angle

Parameter 4 Name	Meaning of parameter 4	Parameter 5 Name	Meaning of parameter 5	Parameter 6 Name	Meaning of parameter 6
IndexFinger	index finger pose angle	ThumbUpFinger	upper thumb pose angle	ThumbDownFinger	Posture angle of thumb lower segment

5.2.3.2.3 Set One Hand Pose Set Fingerpos

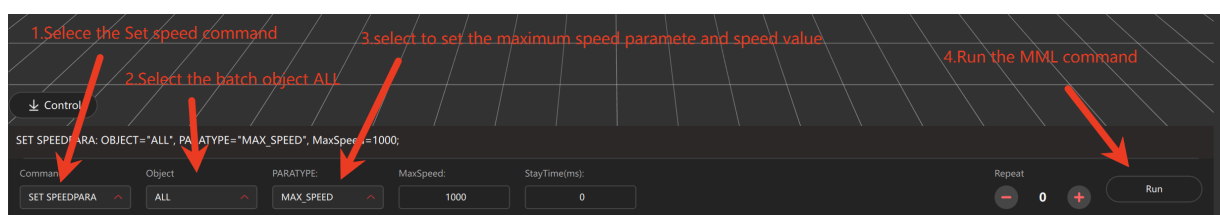
The single-finger pose command enables individual control of manipulator digits to achieve specified angular configurations. The following pose parameters are supported:

FINGER	function	name of parameter	parameter meaning
LITTLE_FINGER	Set Pinkie Pose	Position	Pinkie pose angle
RING_FINGER	Set ring finger pose	Position	Angle of ring finger pose
MIDDLE_FINGER	Set middle finger pose	Position	middle finger pose angle
INDEX_FINGER	Set index finger pose	Position	index finger pose angle
THUMB_UP	Set the upper thumb position	Position	upper thumb pose angle
THUMB_DOWN	Set the lower thumb position	Position	Posture angle of thumb lower segment

5.2.3.3 Batch Object Mode

ALL joint motors and manipulators support operation and control of all joint motors and manipulators. The OBJECT type is "ALL".

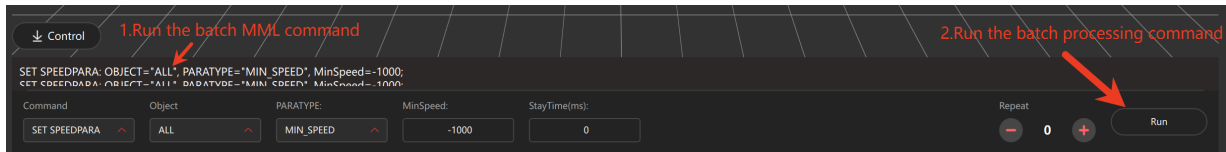
For example, SET COMMAND to "SET SPEEDPARA", OBJECT to "ALL", PARATYPE to "MAX_SPEED", and MaxSpeed to 1000 to set the maximum speed of all joint motors.



5.2.3.4 Batch MML Command Mode

The MML command editor enables batch control functionality through multi-command input. To simultaneously configure maximum/minimum speed parameters for all joint motors, enter the following commands in the MML command text field:

```
SET SPEEDPARA: OBJECT="ALL", PARATYPE="MAX_SPEED", MaxSpeed=1000;
SET SPEEDPARA: OBJECT="ALL", PARATYPE="MIN_SPEED", MinSpeed=-1000;
```



5.2.3.5 Delay Mode

The delay mode controls the wait time for executing MML commands. There are two ways to achieve this delay:

1. Run the StayTime parameter on the end of the MML command to set the delay after the MML command is executed (unit: ms).

```
SET MOTIONPARA: OBJECT="Joint0", PARATYPE="ANGLE", Angle=100, StayTime=1000;
```

2. Use the independent MML command "SET SUSPENDTIME" to specify the delay duration of the StayTime parameter for a specific joint motor/manipulator. The command is:

```
SET SUSPENDTIME: OBJECT="Joint0", PARATYPE="BY_TIME", StayTime=1000;
```

5.2.3.6 Repeat Pattern

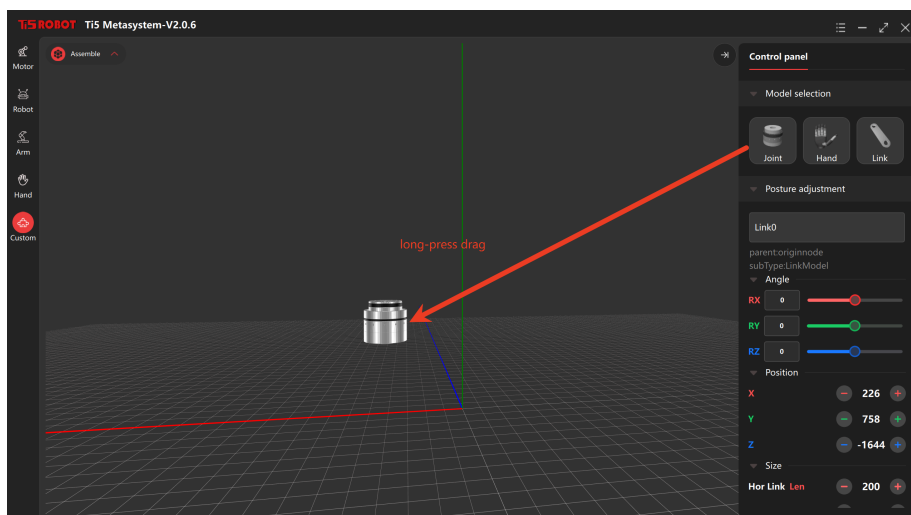
The repeat mode controls the iteration count for MML batch command execution. By specifying a non-zero value for the Repeat parameter, the system will reprocess the batch commands for the designated number of cycles.

5.3 Give A Typical Example

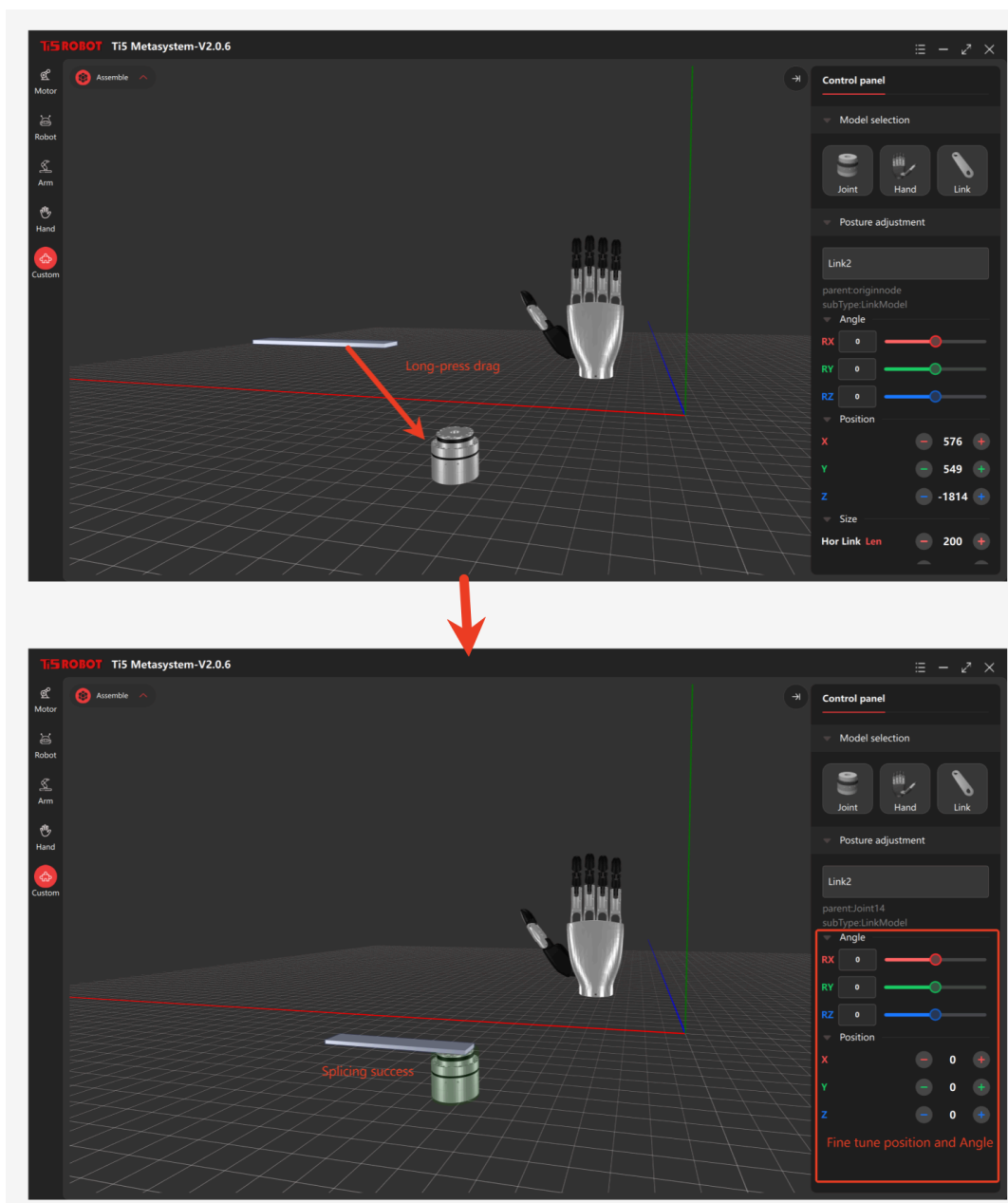
5.3.1 Building Models

5.3.1.1 Drag Generation Aand Connection

After entering construction mode, press and drag the icon from the right material library to the main view area to generate the model, as shown in the figure below.

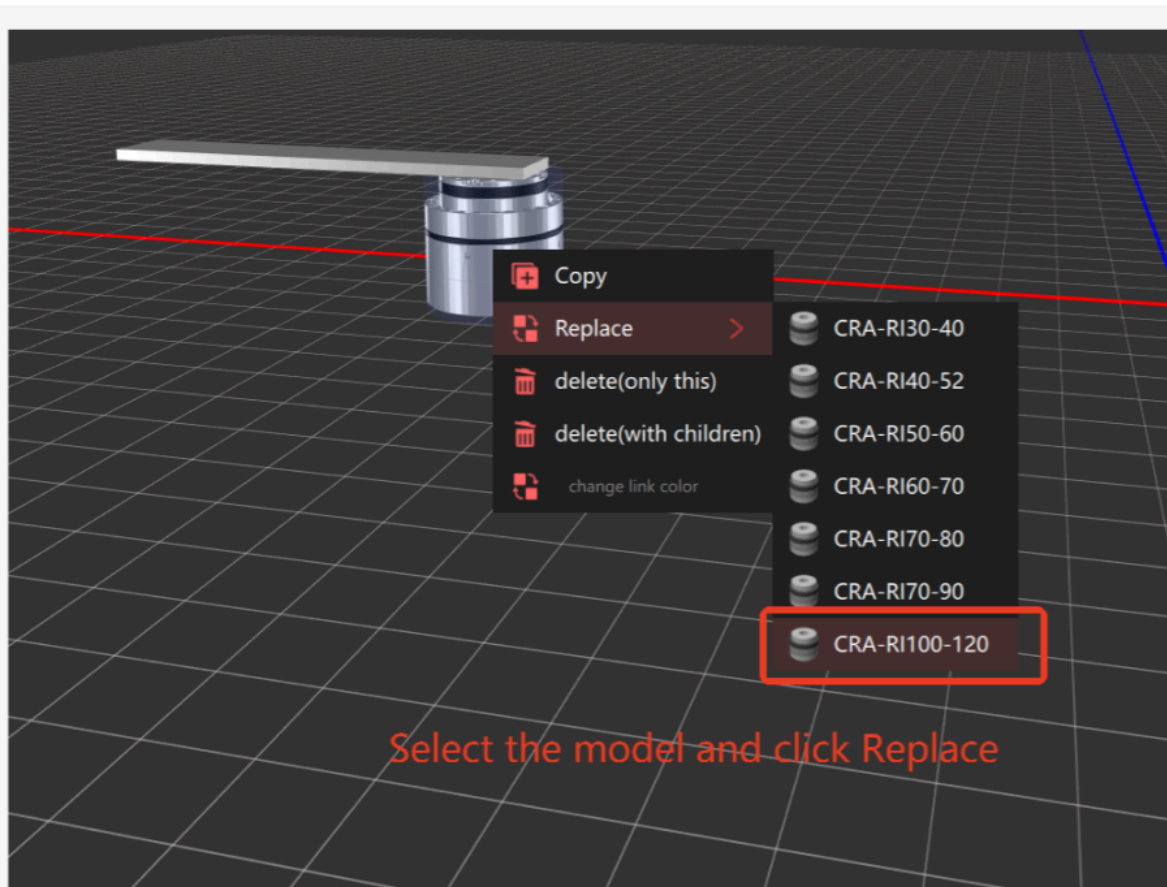


Use the same method to create the manipulator and connecting rod. Press and hold the connecting rod to display the blue state. Drag it to other models (such as the motor in the picture), and the target motor will display the green state. Let go to realize the splicing. When stitching is complete, fine-tune the position and angle on the right at Control Bar => Posture Adjustment.

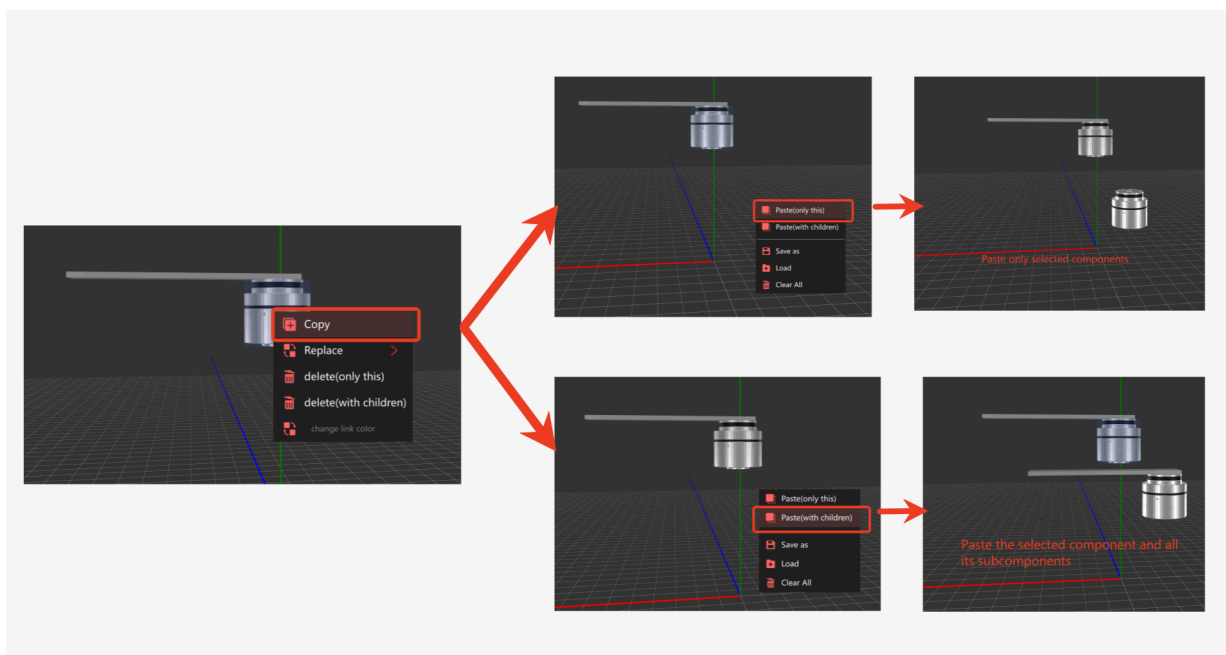


5.3.1.2 Basic Model Operations (Replace, Copy, Delete)

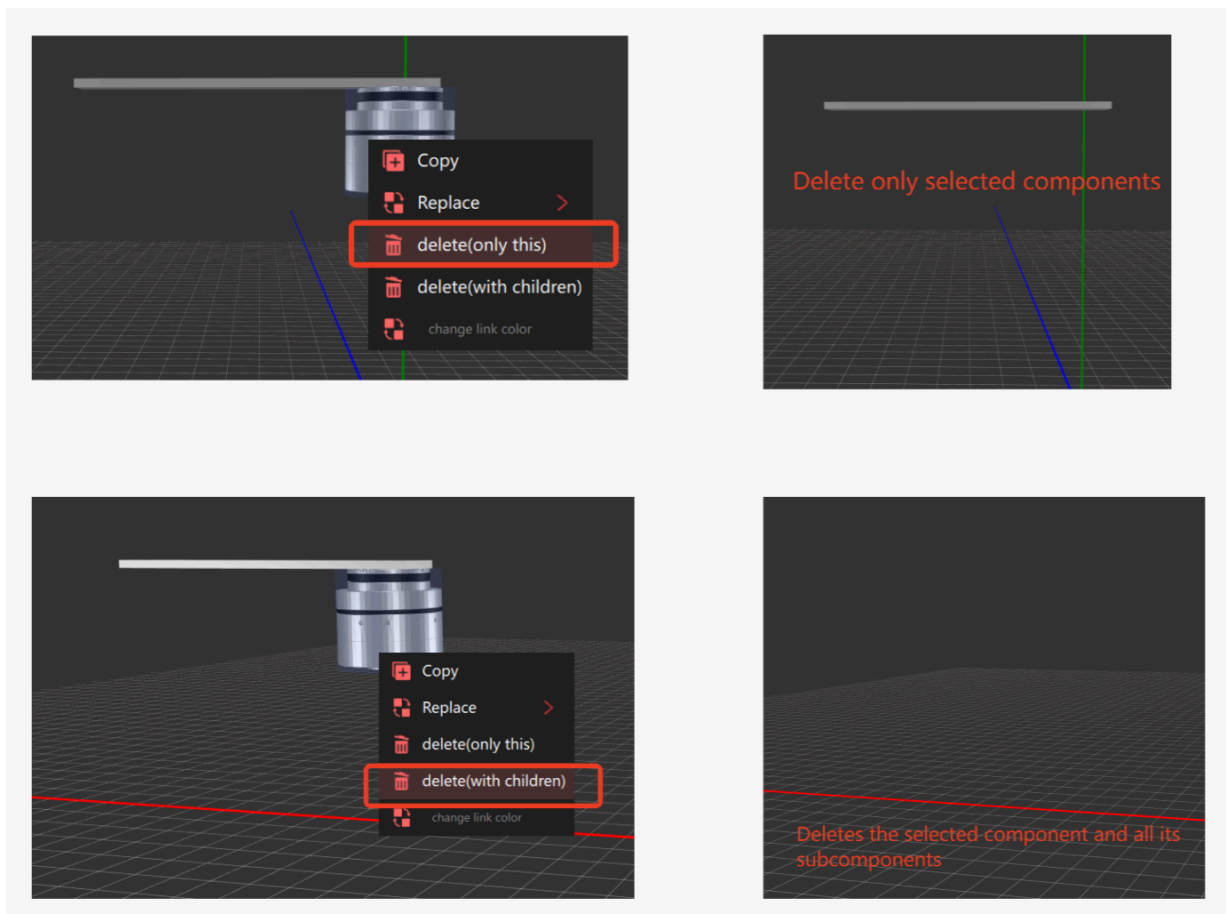
Right-click on the model and select Replace to replace it.



Right-click on the model to call out the menu bar to perform Copy & Paste operation, as shown in the figure below.

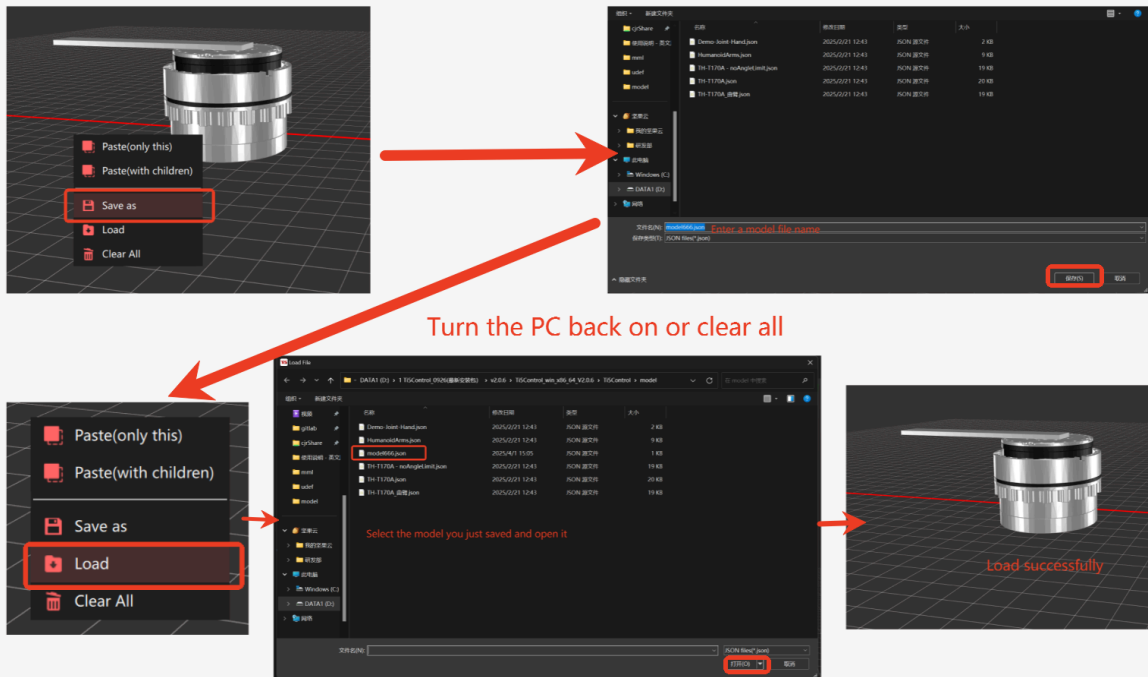


Right-click the model and select "Delete" to delete the model. The logic of "only itself" and "whole chain" is similar to "copy & paste", as shown in the figure.

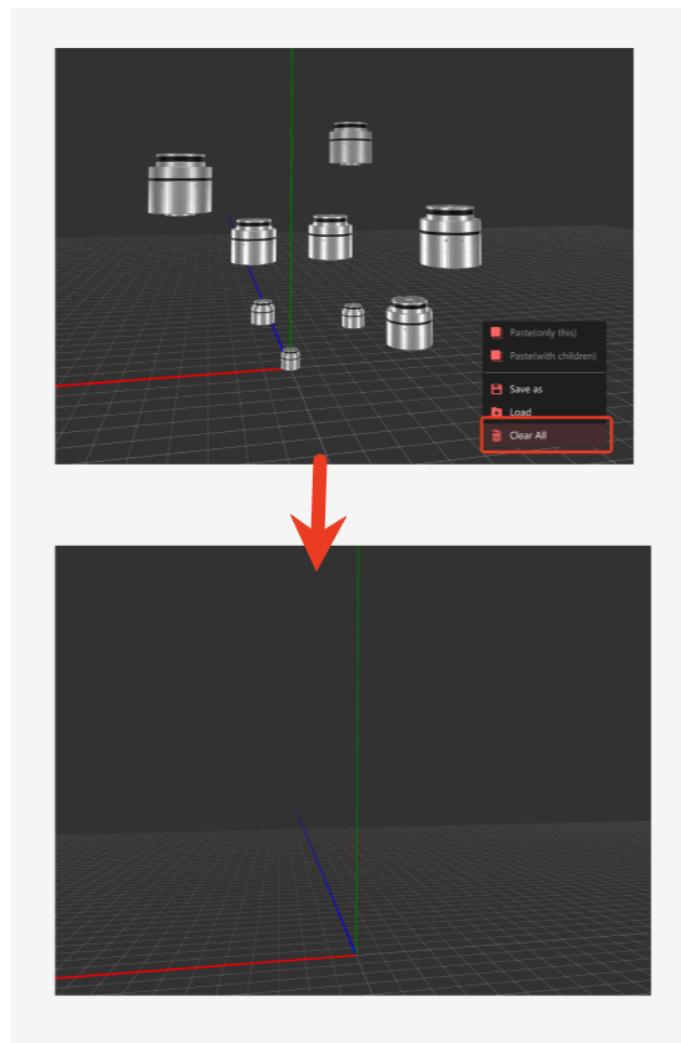


5.3.1.3 Saving And Loading Models

Right-click the blank space to find the "Save As" and "Load" functions, we can store the built model for the next use, or load some of the preset models provided by the official.

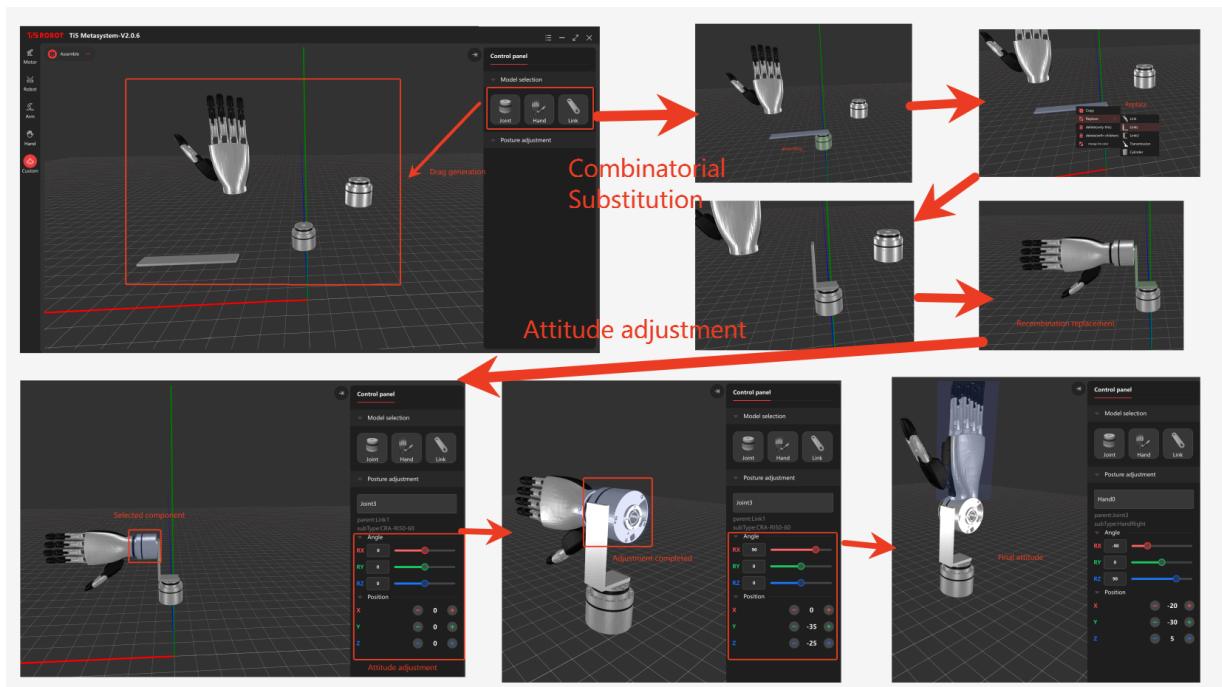


Right click on the blank and select Clear All to delete all models.

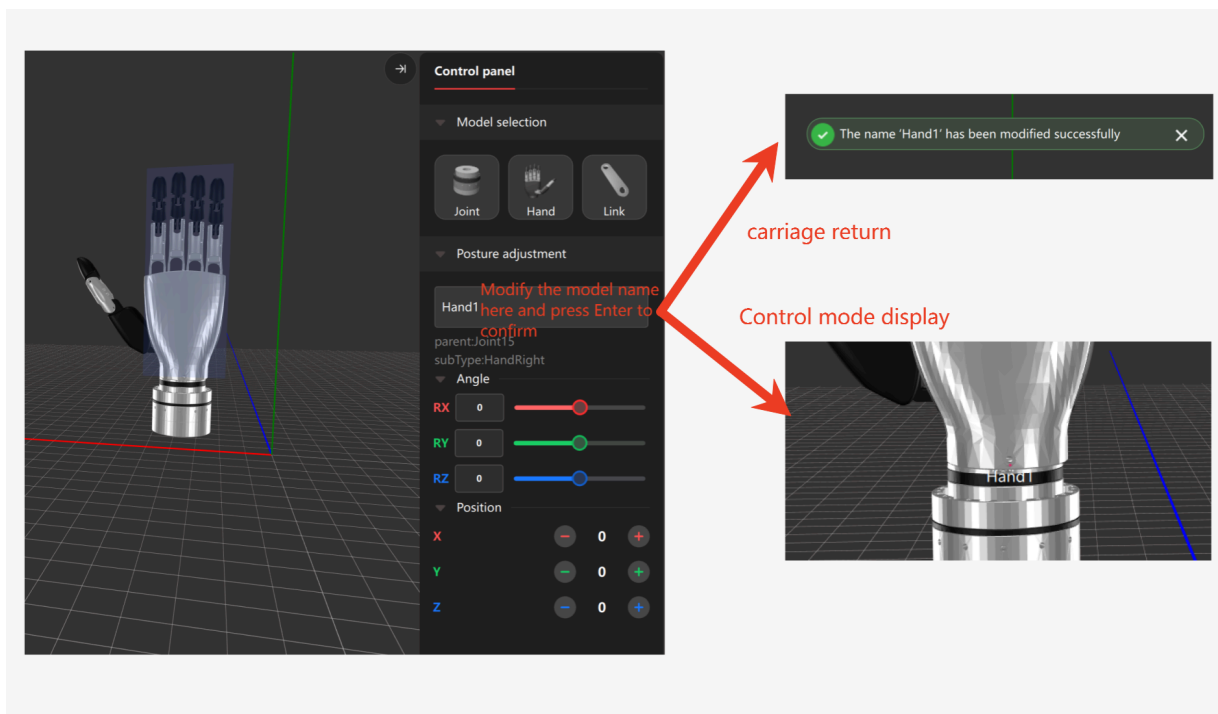


5.3.1.4 Model Attitude Adjustment

If you want to build a normal robot, you need to use the posture adjustment module on the right control bar. As shown in the figure, you can build a waving robot by model combination replacement and posture adjustment.



The model can be named in the posture adjustment bar, and the model name will be displayed in the control mode.

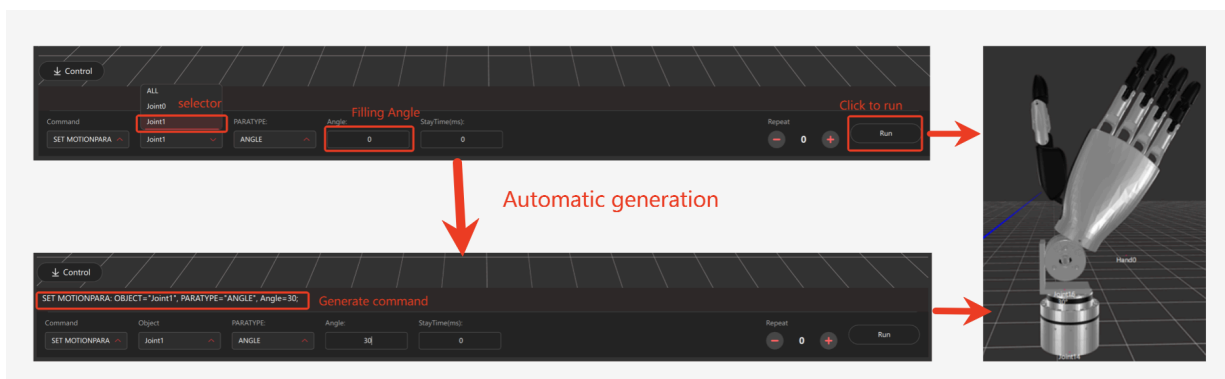


5.3.2 Control Model

Enter control mode, you can use command control function to let the robot wave.

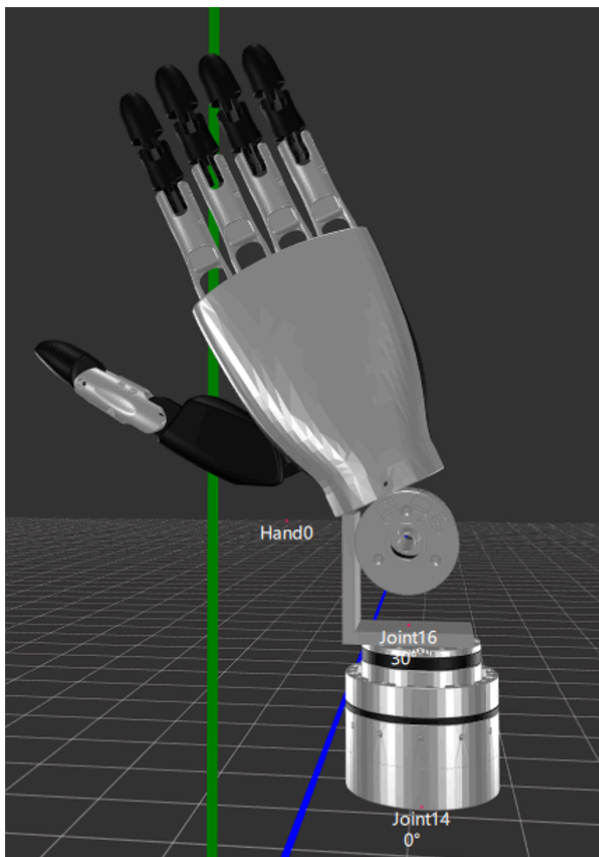
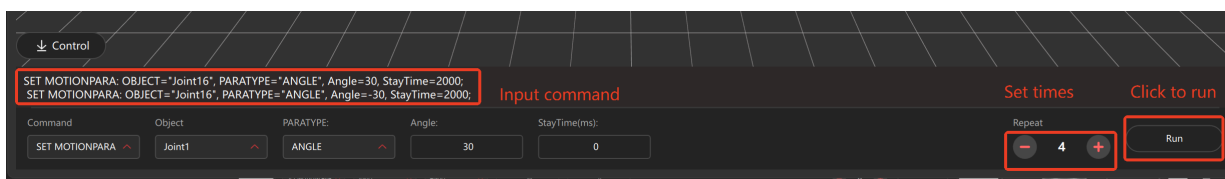
5.3.2.1 Robot Beckons

Find the command area below, first select the object to control, then fill in the angle, click Run, you can see the waving robot successfully "waving". You can see that the command input area above generates an equivalent command at the same time, and we can also use the command to control it.



Mastering command control mode can realize more complex actions. Enter the following command, set the number of repetitions, click Run, you can see our waving robot start waving around.

```
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=30, StayTime=2000;
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=-30, StayTime=2000;
// Refer to commands. For more commands, see 2.3 Command Control.
```



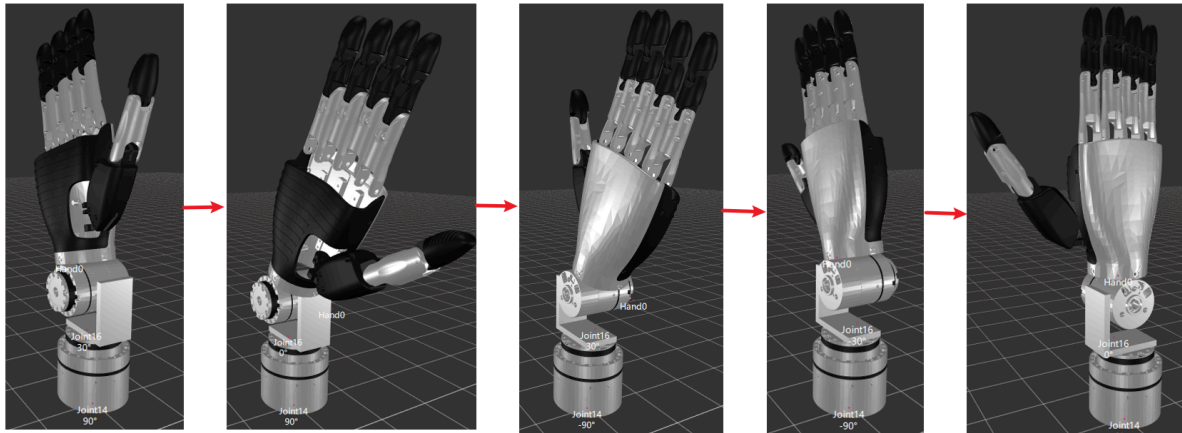
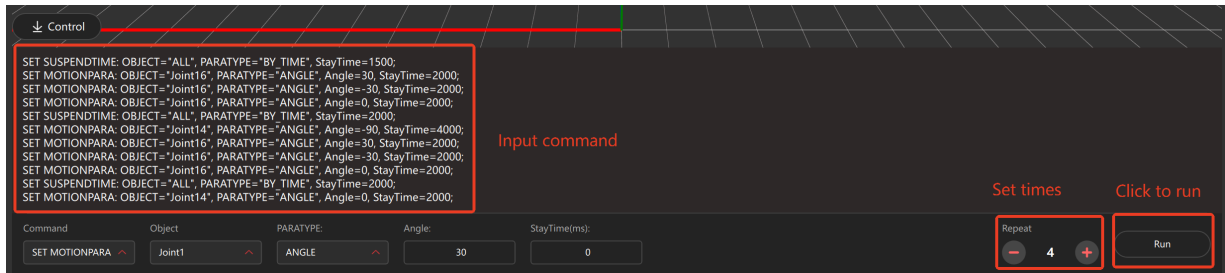
5.3.2.2 Robot Waving Head Waving

Next, try a more complicated logic-swing your head left and right and wave, so that the robot first turns left, then waves, then turns right, then waves, with the following commands.


```

SET MOTIONPARA: OBJECT="Joint14", PARATYPE="ANGLE", Angle=90, StayTime=4000;
SET SUSPENDTIME: OBJECT="ALL", PARATYPE="BY_TIME", StayTime=1500;
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=30, StayTime=2000;
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=-30, StayTime=2000;
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=0, StayTime=2000;
SET SUSPENDTIME: OBJECT="ALL", PARATYPE="BY_TIME", StayTime=2000;
SET MOTIONPARA: OBJECT="Joint14", PARATYPE="ANGLE", Angle=-90, StayTime=4000;
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=30, StayTime=2000;
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=-30, StayTime=2000;
SET MOTIONPARA: OBJECT="Joint16", PARATYPE="ANGLE", Angle=0, StayTime=2000;
SET SUSPENDTIME: OBJECT="ALL", PARATYPE="BY_TIME", StayTime=2000;
SET MOTIONPARA: OBJECT="Joint14", PARATYPE="ANGLE", Angle=0, StayTime=2000;
// Refer to commands. For more commands, see 2.3 Command Control.

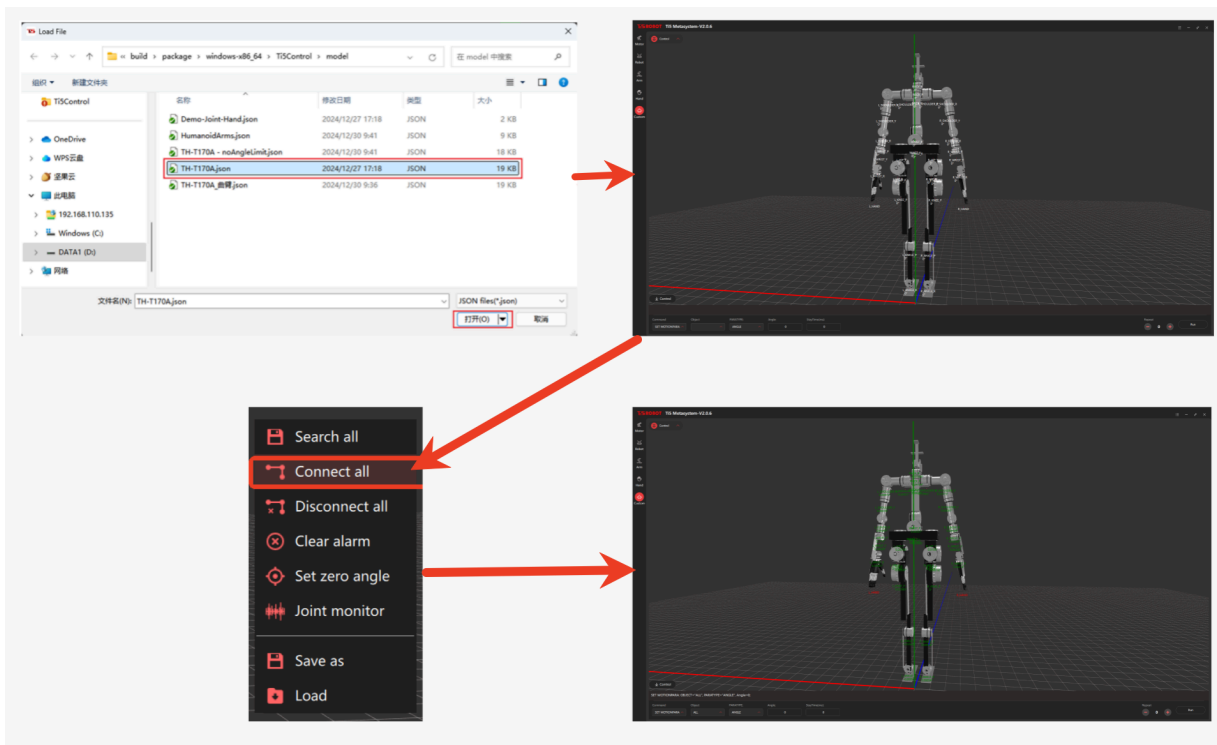
```



5.3.3 Robot 170A Control Case

This section will show how to control large robots by building models. If you have a robot called "170A", you can use the official model instead of building it yourself.

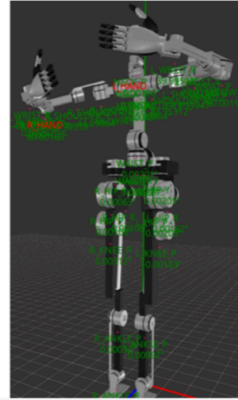
As shown below, directly load the file "TH-T70A.json" provided by us in the model folder, you can see that after connection, the model automatically synchronizes the robot's motor angle (left leg bent inward).



Control large robots through command combinations, the effect is as shown below.

```
// Make a left hand bow
SET MOTIONPARA: OBJECT="L_SHOULDER_P", PARATYPE="ANGLE", Angle=-75;
SET MOTIONPARA: OBJECT="L_SHOULDER_R", PARATYPE="ANGLE", Angle=45;
SET MOTIONPARA: OBJECT="L_SHOULDER_Y", PARATYPE="ANGLE", Angle=-20;
SET MOTIONPARA: OBJECT="L_ELBOW_R", PARATYPE="ANGLE", Angle=-80;
SET MOTIONPARA: OBJECT="L_WRIST_P", PARATYPE="ANGLE", Angle=0;
SET MOTIONPARA: OBJECT="L_WRIST_Y", PARATYPE="ANGLE", Angle=0;
SET MOTIONPARA: OBJECT="L_WRIST_R", PARATYPE="ANGLE", Angle=40;
SET HANDACTION: OBJECT="L_HAND", ACTION="OPEN";

// Make a bow with the right hand
SET MOTIONPARA: OBJECT="R_SHOULDER_P", PARATYPE="ANGLE", Angle=65;
SET MOTIONPARA: OBJECT="R_SHOULDER_R", PARATYPE="ANGLE", Angle=-45;
SET MOTIONPARA: OBJECT="R_SHOULDER_Y", PARATYPE="ANGLE", Angle=20;
SET MOTIONPARA: OBJECT="R_ELBOW_R", PARATYPE="ANGLE", Angle=80;
SET MOTIONPARA: OBJECT="R_WRIST_P", PARATYPE="ANGLE", Angle=0;
SET MOTIONPARA: OBJECT="R_WRIST_Y", PARATYPE="ANGLE", Angle=0;
SET MOTIONPARA: OBJECT="R_WRIST_R", PARATYPE="ANGLE", Angle=-40;
// Refer to commands. For more commands, see 2.3 Command Control.
```



Control

```
SET MOTIONPARA: OBJECT="L SHOULDER_P", PARATYPE="ANGLE", Angle=-75;  
SET MOTIONPARA: OBJECT="L SHOULDER_R", PARATYPE="ANGLE", Angle=45;  
SET MOTIONPARA: OBJECT="L SHOULDER_Y", PARATYPE="ANGLE", Angle=-20;  
SET MOTIONPARA: OBJECT="L ELBOW_R", PARATYPE="ANGLE", Angle=-80;  
SET MOTIONPARA: OBJECT="L WRIST_P", PARATYPE="ANGLE", Angle=0;  
SET MOTIONPARA: OBJECT="L WRIST_Y", PARATYPE="ANGLE", Angle=0;  
SET MOTIONPARA: OBJECT="L WRIST_R", PARATYPE="ANGLE", Angle=40;  
SET HANDACTION: OBJECT="L HAND", ACTION="OPEN";  
  
SET MOTIONPARA: OBJECT="R SHOULDER_P", PARATYPE="ANGLE", Angle=65;  
SET MOTIONPARA: OBJECT="R SHOULDER_R", PARATYPE="ANGLE", Angle=-45;  
SET MOTIONPARA: OBJECT="R SHOULDER_Y", PARATYPE="ANGLE", Angle=20;  
SET MOTIONPARA: OBJECT="R ELBOW_R", PARATYPE="ANGLE", Angle=80;  
SET MOTIONPARA: OBJECT="R WRIST_P", PARATYPE="ANGLE", Angle=0;  
SET MOTIONPARA: OBJECT="R WRIST_Y", PARATYPE="ANGLE", Angle=0;  
SET MOTIONPARA: OBJECT="R WRIST_R", PARATYPE="ANGLE", Angle=-40;
```

Command	Object	PARATYPE	Angle	StayTime(ms)
SET MOTIONPARA		ANGLE	0	0

Repeat

-

4

+

Run